Overlapping Generations in Lean: The Life-Cycle Household

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Abstract

This is the record of a programme to formalise overlapping-generations economies in Lean 4, and of one result it reaches. In Diamond's two-period economy the steady state is unique for every relative risk aversion and every depreciation rate, with no condition on the savings function, even though the saving rate itself falls with the return whenever risk aversion exceeds one: the wage channel of a constant-returns technology dominates. That is worth stating first because it shows a uniqueness proof need not pass through the household-side comparative static at all. A household living J periods against uninsurable earnings risk and a borrowing constraint solves a finite backward induction, so everything proved for the infinite-horizon Bellman operator with an arbitrary continuation applies stage by stage with no fixed point; on top of that we prove the finite-horizon sandwich $\kappa_k m \le c_k \le \kappa_k(m + H_k(z))$, that κ_k is exactly the reciprocal of $\sum_{i\le k}(\text{þ}/R)^i$, and that the propensity to consume is at least κ_k — the companion of the upper bound on its slope, and the piece the uniqueness argument turns out to need. Adding retirement and a pension settles a claim often made informally: a means test does make the policy non-monotone, but not for the reason usually given, since saving rises with cash on hand whatever the continuation, by Topkis, so a non-concave value function cannot by itself reverse it. A stationary equilibrium exists with both boundary conditions discharged from primitives: capital supply is a finite composition, needing no stationary distribution, no Feller property and no Doeblin condition, and it is bracketed by affine bounds on a cohort's accumulated assets whose coefficients are explicit recursions in the primitives.

The equilibrium is unique at logarithmic utility. Single crossing does not need capital supply to rise with the rate, only to fall more slowly than the firm's demand, and that increment is controlled by an envelope whose per-stage multiplier is damped by the propensity floor; the mass weighting in it is exact, because a cohort's earnings marginal is the invariant law at every age; and the multiplier itself is one plus the variance of the risk tilt, which we certify below what the damping allows, in exact rational arithmetic, at every one of the sixty stages and over the whole asset range. Theorem 1 then holds

*The Lean development described here is at `https://github.com/LeanEconomics/LeanEconomics`. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington); every result reported was checked by the Lean kernel. This is the companion to the write-up on Aiyagari (1994) in the same repository. Comments welcome.

at every state and every age, capital supply rises with the rate, and the equilibrium is unique. That certificate has since been superseded: Light's induction on the stage values proves Theorem 1 at every state and age for relative risk aversion at most one, at every calibration, by proof term; the certificate stands as the record of the only route that reaches above one, and the sharp moment bound shows that route stops at $\gamma \approx 1.5$.

Above logarithmic utility it is open, and the paper is as much a record of where it is blocked. Two routes are closed with quantified reasons. Date-by-date Lasry–Lions monotonicity fails in this economy at every risk aversion including logarithmic, and not because the price channel runs backwards — it is exactly the congestion term — but because strict concavity reweights it; the summed condition that the argument actually uses does hold, but its weakest form collapses to monotone capital supply, so the mean field route is not weaker than the single crossing it was meant to bypass. And the certainty-equivalent criterion that carries the logarithmic case fails from $\gamma \approx 1.5$: the propagation is damped at every risk aversion we test, and it is the certainty-equivalent side that gives way.

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1 Introduction

The companion write-up in this repository proved that the stationary equilibrium of Aiyagari (1994) is unique at his logarithmic calibration, and located exactly what stands in the way for relative risk aversion $\mu \in \{3, 5\}$. Every obstacle it met traces to the fixed point of the infinite-horizon problem: a condition on the response of consumption to the interest rate fails in a tail of wealth that the stationary distribution reaches; an induction that would prove the response bound has a step multiplier $1/\text{þ} > 1$, so it cannot close as a supremum argument; and the only floor on capital supply comes from an ergodic identity that loses a factor $\|u'\|^2/\operatorname{Var}$. Overlapping generations remove all three at once. A household that lives J periods solves a finite backward induction from an explicit terminal condition, so a multiplier $1/\text{þ}$ per step compounds to þ^{-J}, which at Aiyagari's numbers is about 1.03. Wealth is bounded by J periods of accumulation, so any condition needs checking on a known bounded region. And the distribution of the youngest cohort is common to every interest rate, so an ordering of policies pushes forward to an ordering of the whole age profile of distributions by induction on age, with no Doeblin atom and no uniqueness of a stationary distribution.

The target of the programme is uniqueness of the stationary equilibrium of a stochastic life-cycle economy in the manner of Huggett (1996) at $\mu \in \{3, 5\}$. It is reached at $\mu = 1$ and not above, and the paper is as much a record of where it is blocked as of what is proved.

Sections 2 to 6 are the foundations. Section 2 does Diamond's two-period economy, where the steady state is unique for every risk aversion and the household-side comparative static that blocks the general case runs the wrong way without doing any harm. Section 3 sets out the life-cycle household, the finite-horizon sandwich, and the propensity floor. Section 5 adds retirement, a pension and a means test, and settles which policy a means test makes

non-monotone and in which state space. Section 6 proves existence of a stationary equilibrium with both boundary conditions discharged from primitives rather than assumed: capital supply in a life cycle is a finite composition of finite sums, so existence needs no stationary distribution and no compactness argument beyond Berge, and it can be bracketed by two affine bounds on a cohort's accumulated assets sharing a slope, with every coefficient an explicit recursion in the primitives. At Aiyagari's calibration the hypotheses hold on an interval of about five percentage points containing the true rate of 5.09%.

Sections 7 to 10 are about uniqueness, and they are best read as three attempts on one obstacle. Uniqueness needs capital supply to rise with the interest rate, and the only route the literature offers is Light's Theorem 1, that saving rises with the rate at every state. Section 7 reports how far that goes: it holds at every age below a critical relative risk aversion of about 3.4, so the life cycle covers Aiyagari's $\mu = 3$ and not his $\mu = 5$; the Euler reduction of it is false at logarithmic utility exactly where an equilibrium sits, and a different sufficient condition, on the value gap between two rates, is true there instead (Section 8). The section also separates the mechanism from the usual story: the reversal happens in the certainty-equivalent household, which faces no risk at all, so precautionary saving is not what drives it, and we give the criterion in closed form and prove its certainty-equivalent half for every risk aversion.

Section 9 asks what mean field games offer. The Lasry–Lions argument was dismissed for the stationary Bewley model because it needs a common initial distribution, which overlapping generations have; so the monotonicity condition becomes a question to check rather than one to dismiss. Date by date it fails, at every risk aversion, and not for the expected reason. What the argument actually uses is weaker — only the summed pairing is signed — and that weaker condition holds throughout the bracket in which the equilibrium is known to lie. But pushed to its weakest form it collapses to monotone capital supply, so the route is not logically weaker than the one it was meant to bypass. The section is a closure, with the reduction and the two theorems it produces kept because they are about the life-cycle household rather than about mean field games.

Section 10 is the route that works at $\mu = 1$. Single crossing does not ask capital supply to rise, only to fall more slowly than the firm's demand; over the proved bracket the demand curve is steep enough that a proof may lose more than the true increment and still close. Coupling the two economies on a common earnings path and splitting the step at the lower-rate state gives an envelope in which the failure of Theorem 1 is priced rather than excluded, and its per-stage multiplier is damped by the propensity floor. Two further facts make the estimate small enough: a cohort's earnings marginal is the invariant law at every age, so the mass weighting is exact in the coordinate where the failure lives; and the multiplier is one plus the variance of the risk tilt, an identity rather than a bound. That variance is certified below what the damping allows, in exact rational arithmetic, at every stage and over the whole asset range. What that delivers is Theorem 1 at every state and age, hence monotone capital supply, hence uniqueness — at logarithmic utility, at one calibration, with the last hypothesis discharged by certificate rather than by proof term. Section 8.3 then removes the

certificate: Light's induction on the stage values gives Theorem 1 at every state and age for $\gamma \leq 1$ at every calibration, by proof term. Section 11 records what remains.

As in the companion, every result stated with a formal name has been checked by the Lean kernel against Mathlib; the development is at `https://github.com/LeanEconomics/LeanEconomics`, directory `LeanEconomics/OLG`, and the numerical work is in `WriteUps/WriteUpOLG/numerics`.

2 Diamond's two-period economy

The base case of the programme, and the one place in it where uniqueness is proved outright rather than reduced to an assumption. Agents live two periods, work when young and consume out of savings when old; a representative firm rents capital from the old.

2.1 The household, in closed form

A young household with wage w facing gross return R solves $\max u(w-s) + \beta u(Rs)$. With CRRA utility the Euler equation $u'(w-s) = \beta R\, u'(Rs)$ has the explicit solution $s = \sigma(R)w$ with

$$\sigma(R) = \frac{1}{1 + \beta^{-1/\gamma} R^{(\gamma-1)/\gamma}}$$

(`saveShare`). That σ satisfies the Euler equation is algebra (`saveShare_euler`); that it is the optimum needs one more ingredient.

Proposition 2.1 (CRRA lies below its tangent)**.** *For $c, c' > 0$ and $\gamma > 0$, $u(c') \leq u(c) + u'(c)(c'-c)$.*
Formal statement: *crra_le_tangent.*

This is Bernoulli's inequality in both directions. Below unit risk aversion the exponent $1-\gamma$ lies in $(0,1)$ and the concave form applies directly; above it the exponent is negative, and inverting the consumption ratio and multiplying through turns the claim into the form with exponent $\gamma \geq 1$. Optimality then follows by adding the tangent inequality at the two consumption levels and cancelling with the Euler equation, with no differentiation of a value function anywhere (`saveShare_isOptimal`).

2.2 The comparative static is discouraging

Proposition 2.2 (The textbook comparative static)**.** *The saving rate rises with the gross return if $\gamma < 1$ and falls if $\gamma > 1$.*
Formal statement: *saveShare_strictMonoOn.*

, saveShare_strictAntiOn

So the household-side monotonicity that Section 7 cannot establish in general is here a one-line calculation, and it runs the *wrong* way at the risk aversions economists use: above unit intertemporal elasticity the income effect beats the substitution effect and a household facing a better return saves a smaller share of a given wage.

2.3 The steady state is unique anyway

What rescues it is general equilibrium. With Cobb-Douglas production the wage rises and the return falls as capital deepens, and the wage channel dominates. Writing $k' = \sigma(R(k))\, w(k)$ for capital per worker next period (`diamondMap`), the whole argument is that k'/k falls.

Proposition 2.3 (The engine)**.** *If $\theta \le 1$, $B > 0$, $a \ge 0$ and $b > 0$, then $x \mapsto x/(1+B(a+bx)^\theta)$ is strictly increasing on $(0, \infty)$.*
Formal statement: *`div_one_add_rpow_strictMono`.*

The proof is the chain $x_2/x_1 \le R_2/R_1 \le (R_2/R_1)^\theta$, where the first step is $ax_2 \le ax_1$ and the second is that a ratio at most one rises when raised to a power at most one. No calculus. Substituting $x = k^{\alpha-1}$, which falls with k, and $\theta = (\gamma-1)/\gamma$, which is below one for every positive γ, gives the result.

Theorem 2.4 (Existence and uniqueness)**.** *For every $\gamma > 0$, every $\delta \le 1$ and Cobb-Douglas production, the map $k \mapsto k'/k$ is strictly decreasing, so there is at most one positive steady state; and if the map crosses the diagonal between two positive levels there is exactly one.*
Formal statement: *`exists_unique_steadyState`.*

`, steadyState_unique, exists_steadyState`

No condition on the savings function is needed, of the kind textbook treatments impose. The numerical check is in `numerics/dcheck.m`: the closed form matches a grid search to within the grid spacing at every risk aversion and return tried; there is exactly one positive crossing of the diagonal for risk aversions from 0.5 to 10 at both $\delta = 1$ and $\delta = 1/2$; and the saving rate rises with the return only below unit risk aversion, as Proposition 2.2 says.

What this does and does not settle is worth stating. It settles uniqueness in the two-period economy completely, for every risk aversion, which the stochastic life cycle of Section 6 does not. It settles it by an argument that has no counterpart there, because with more than two periods the household's saving is not a closed form and k'/k is not a product of two monotone factors. What it does supply is the shape of the answer: the household-side comparative static can fail while the general-equilibrium one holds, so a proof of uniqueness need not go through Light's Theorem 1 at all.

3 The life-cycle household

3.1 Stages

The household of the companion write-up is kept: earnings $y(z)$ follow a finite Markov chain π, the gross return is $R = 1 + r$, cash on hand is $m = Ra + y(z)$, saving $a' \in [0, \bar{a}]$ is bounded below by the borrowing limit and above by an artefactual cap, utility is CRRA with $u'(c) = c^{-\mu}$ and $\beta \in (0, 1)$. What changes is the horizon: the household lives J periods and leaves nothing behind. Count periods from the end. With k periods still to come after today, the continuation value is the k-th Bellman iterate from zero,

$$V_k = T^k 0, \qquad (Tv)(a, z) = \max_{a'} \{u(m - a') + \beta \textstyle\sum_{z'} \pi(z, z') v(a', z')\},$$

and the household with k periods to come saves $g_k = \text{policyOf}\, V_k$ and consumes $c_k = m - g_k$ (`stageValue`, `stagePolicy`, `stageConsumption`). Age j of a J-period life is stage $k = J - 1 - j$. In the last period the household saves nothing and consumes its cash on hand (`stagePolicy_zero`, `stageConsumption_zero`).

This is the finite-horizon problem with a flat age profile of earnings: the same income process at every age. An age profile, retirement and survival risk need income to be a parameter of the Bellman step rather than a field of the structure, and are deferred to the next stage of the programme.

3.2 What transfers without a fixed point

The companion development proved its household results for the Bellman operator applied to an arbitrary continuation v, and only then passed to the fixed point. Here there is no fixed point to pass to, and each result is read at $v = V_k$:

- consumption is positive at every stage when marginal utility is unbounded at zero (`stageConsumption_pos`);
- the stage value has concave slices, so the optimal action is unique and both saving and consumption rise with wealth (`stagePolicy_mono`, `stageConsumption_mono`);
- Carroll–Kimball: consumption is concave in wealth at every stage (`concaveOn_stageConsumption`);
- the Euler inequality under the cap, $\beta R\,\mathbb{E}[u'(c_k(a', z')) \mid z] \le u'(c_{k+1}(a, z))$, at every state (`stage_euler_le`), and the Euler inequality above the floor, the reverse, at every state where the household saves (`stage_euler_ge`).

3.3 The finite-horizon sandwich

Let $\mathfrak{p} = (\beta R)^{1/\mu}$ and define the finite-horizon minimal propensities to consume by

$$\kappa_0 = 1, \qquad \kappa_{k+1} = \frac{\kappa_k R}{\mathfrak{p} + \kappa_k R}, \qquad \text{so that} \qquad \frac{1}{\kappa_k} = \sum_{i=0}^{k} \Big(\frac{\mathfrak{p}}{R}\Big)^i$$

(`stageMPC`): the share of cash on hand consumed by a household with k periods to come and no future income. Let $H_0 = 0$ and $RH_{k+1}(z) = \sum_{z'} \pi(z, z')(y(z') + H_k(z'))$, the expected present value of the next k incomes (`stageHumanWealth`).

Theorem 3.1 (The finite-horizon sandwich)**.** *Assume CRRA utility with marginal utility unbounded at zero, and that the cap is slack at every stage. Then for every k, every z and every $a \in [0, \bar{a}]$,*

$$\kappa_k\, m \;\le\; c_k(a, z) \;\le\; \kappa_k (m + H_k(z)).$$

Formal statement: *`IncomeFluctuation.stageMPC_mul_le_stageConsumption`, `IncomeFluctuation.stageConsumption_le_stageHumanWealth`.*

Each bound is one Euler step, applied k times. The lower bound: if consumption against v is at least κm, the Euler inequality above the floor at a state where the household saves gives $c^{-\mu} \leq \beta R\,\mathbb{E}[(\kappa R a')^{-\mu}]$ with $a' = m - c$, hence $\kappa R(m-c) \leq \text{þ} c$ and $c \geq \kappa R(\text{þ} + \kappa R)^{-1} m$; at a corner $c = m$ (`crra_stageMPC_step`). The upper bound: if consumption against v is at most $\kappa(m + H(z))$, the Euler inequality under the cap and Jensen for the negative power give $c^{-\mu} \geq \beta R\,(\mathbb{E}[c'])^{-\mu}$ with $\mathbb{E}[c'] \leq \kappa R(a' + H'(z))$, hence $c \leq \kappa R(\text{þ} + \kappa R)^{-1}(m + H'(z))$ (`crra_stageUpper_step`). The recursion for κ is the same on both sides, which is why the sandwich is exact in the homogeneous case.

The propensities decrease in the horizon towards the infinite-horizon $\kappa = 1 - \text{þ}/R$ of the companion write-up: $\kappa \leq \kappa_k$ for every k (`minMPC_le_stageMPC`), so a household with less life left consumes a larger share of its cash, and the infinite-horizon bounds are the limiting case. A first consequence for the programme is an explicit bound on saving, $g_k(a, z) \leq (1 - \kappa_k)(Ra + y_{\max})$ (`stagePolicy_le`), which iterated from zero wealth at the start of life bounds the wealth of every cohort.

4 The deterministic life cycle

Between Diamond's two periods and the stochastic life cycle sits the case with many periods and no earnings risk, the Auerbach–Kotlikoff class. It is worth doing on its own, because there the sandwich of Theorem 3.1 closes and Light's Theorem 1 becomes algebra.

4.1 The sandwich closes

Make income the same in every state, and suppose the household is interior at every age: it saves something and stops short of the artefactual cap.

Theorem 4.1 (Consumption in closed form)**.** *Under those two conditions, $c_k(a, z) = \kappa_k(m + H_k)$ exactly, so saving is exactly $(1 - \kappa_k)(y + Ra) - \kappa_k H_k$.*
Formal statement: *IncomeFluctuation.stageConsumption_eq_of_interior.*

The proof squeezes the Euler equation between its two directions. Interiority gives both the inequality above the floor and the one under the cap, hence equality; constant income makes tomorrow's consumption the same in every state, so the expectation collapses to a point and the power can be inverted; the budget identity closes the induction. What the stochastic model can only bracket is here an identity.

4.2 Uniqueness at logarithmic utility

Theorem 4.2 (Theorem 1, deterministic and logarithmic)**.** *At $\gamma = 1$, saving rises with the interest rate at every age and every asset level, with no condition. Aggregate capital rises with it, and the stationary equilibrium rate is unique.*
Formal statement: *IncomeFluctuation.stagePolicy_mono_det_log.*

, IncomeFluctuation.olgCapital_mono_det_log

Two facts do it. At $\gamma = 1$ the patience factor is βR, so the R cancels out of the recursion for κ and the propensity does not depend on the rate at all (`stageMPC_one_eq_of_rate`); and human wealth falls as the rate rises (`stageHumanWealth_antitone_rate`). The gap is then exactly

$$g(R_2) - g(R_1) = (1 - \kappa_k)(R_2 - R_1)a + \kappa_k\big(H_k(R_1) - H_k(R_2)\big),$$

and both terms are nonnegative. Compare Section 8, where at logarithmic utility with risk the Euler condition is *false* at the rates that matter and a different sufficient condition has to be found. Risk is what makes it hard, not the horizon.

4.3 Above logarithmic utility

For $\gamma \neq 1$ the propensity moves with the rate and the comparative static becomes a duration comparison. Consumption at zero assets is $y \sum_{i \le K} R^{-i} / \sum_{i \le K} q^i$ with $q = \text{Þ}/R$, and it falls with the rate exactly when

$$T_y \;\ge\; \Big(1 - \tfrac{1}{\gamma}\Big) T_c,$$

where T_y and T_c are the durations of the income and consumption profiles. Positive assets only help, since they add $(1 - \kappa_k)(R_2 - R_1)a \ge 0$.

It is worth being careful about what that threshold is. The certainty-equivalent analysis of Section 7 reports a critical risk aversion of 4.33, and it is tempting to read that as the deterministic threshold. It is not. That number is the certainty equivalent of the *risky* problem read at the state that matters, where expected income falls back towards the mean and so is front-loaded relative to consumption. With a flat income profile the two durations are nearly equal, and the criterion has enormous room (`numerics/detcheck.m`):

r	0%	2%	4%	6%
Income duration T_y	29.50	23.70	18.70	14.79
Consumption duration T_c	28.89	23.40	18.68	14.95
T_y / T_c	1.021	1.013	1.001	0.989
Criterion binds at γ of	never	never	never	94

Below the rate of time preference the income profile is in fact the *longer* of the two, so $T_y > T_c$ and the criterion holds for every γ whatever; above it consumption is the longer, and the criterion first fails at a risk aversion near 94. Either way there is no threshold an economist would meet: the derivative of saving with respect to the rate at $r = 4\%$ is positive and comfortably so from $\gamma = 1$ to $\gamma = 5$, at zero assets and at five units of them. The threshold of Section 7 is a feature of the earnings *profile* the risky problem induces, not of risk aversion by itself. Only the logarithmic case is formalised, because above it the duration comparison is an inequality between two geometric sums with different ratios, which is true here but not unconditionally.

5 Retirement, a pension, and the means test

5.1 Retirement with a lump-sum pension

The household works until age J_r and is retired after it, receiving a pension in place of its labour earnings. With a *lump-sum* pension p the retired household faces the same problem as the working one with income replaced by the constant p, so the retired phase is itself an income-fluctuation problem (`withPension`) and everything in Section 3 applies to it unchanged. The life-cycle value is assembled by iterating the working Bellman operator on the retired value: lifeValue(n_r, n_w) is the value of a household with n_w working and n_r retired periods still to come (`lifeValue`), so a household of age $j < J_r$ in a life of J periods sits at $n_w = J_r - j$ and $n_r = J - J_r$. Saving and consumption rise with wealth at every age (`lifePolicy_mono`, `lifeConsumption_mono`).

Two facts about the pension itself are worth recording. A retiree's human wealth is the same in every earnings state, because the pension does not depend on the earnings shock (`stageHumanWealth_withPension_const`), and it never exceeds the perpetuity value of the pension, $H_k \le p/r$ (`stageHumanWealth_withPension_le`). With Theorem 3.1 the retiree's consumption therefore satisfies

$$\kappa_k\, m \;\le\; c_k(a) \;\le\; \kappa_k\Big(m + \frac{p}{r}\Big)$$

at every retired age. And cash on hand is strictly increasing in assets (`withPension_resources_strictMono`). Both of these fail once the pension is means-tested.

5.2 Means testing

A pension is means-tested when the amount paid falls with the recipient's assets. We write the schedule as a function $p(\cdot)$ of assets and cash on hand as $m(a) = Ra + p(a)$ (`cash`), and consider three schedules (`lumpSum`, `tapered`, `cliff`): a lump sum; a taper, withdrawn at rate τ per unit of assets above a threshold and never negative; and a cliff, withdrawn entirely above the threshold.

It is often argued that a means test makes the household's policy function non-monotone, and the reason usually given is that the test makes the value function non-concave. The argument as usually given is wrong, and the conclusion is nevertheless true, for a different reason. Three results separate the two.

Theorem 5.1 (Monotonicity in cash on hand is never lost)**.** *Let period utility be strictly concave on the positive half-line and let W be* any *continuation value. If $m_1 < m_2$ and x_i is optimal at m_i, then $x_1 \le x_2$.*
Formal statement: *`OLG.isOptimalSaving_mono`.*

The proof is Topkis. Adding the two optimality inequalities gives $u(m_1-x_1)+u(m_2-x_2) \ge u(m_1 - x_2) + u(m_2 - x_1)$; if $x_2 < x_1$ the two right-hand consumptions are strict convex combinations of the two left-hand ones with the same sum, so strict concavity makes the reverse inequality strict, a contradiction. The continuation enters only through terms that

cancel. So a non-concave value function — which a means test certainly produces — cannot by itself make saving fall with cash on hand, whatever the shape of W.

What a non-concave continuation does produce is a jump, and the jump falls on consumption.

Theorem 5.2 (Consumption falls when wealth rises)**.** *There are a continuation value arising from a means-tested pension and two levels of cash on hand, the second larger, at which optimal consumption is strictly smaller. With log utility, a unit gross return, and a pension of one withdrawn entirely above assets of one: at $m = 6$ the household saves exactly 1 and consumes 5; at $m = 8$ it saves 4 and consumes 4.*
Formal statement: *`OLG.exists_consumption_decrease`.*

At $m = 6$ the household keeps the pension, saving the most it can without losing it; at $m = 8$ the pension is no longer worth keeping, so the household gives it up and saves the proceeds. Saving rises from 1 to 4, as Theorem 5.1 says it must, and since it rises by more than wealth does, consumption falls. Both optima are exact: the objective is log of a quadratic on each side of the threshold, and the comparison is checked by the kernel.

The third result is the genuine non-monotonicity of the saving policy, and it comes from the budget rather than from the value function. A retiree's pension is assessed each period on the assets then held, so the schedule enters cash on hand directly.

Theorem 5.3 (A gentle taper is harmless; a cliff is not)**.** *If $0 \le \tau < R$ then $a \mapsto Ra + p(a)$ is strictly increasing under the taper (`cash_strictMono_tapered`), and so, by Theorem 5.1, saving is monotone in assets (`isOptimalSaving_mono_assets`). Under a cliff it is not: if $a \le \bar{a} < b$ and $R(b - a) < p$ then cash on hand is strictly smaller at b (`cash_lt_cash_of_cliff`), and there are two asset levels, the second larger, at which the household saves strictly less: with a concave continuation, at assets 1 it saves 1/2 and at assets 3/2 only 1/4.*
Formal statement: *`OLG.exists_saving_decrease_in_assets`.*

The continuation in the second half of Theorem 5.3 is $\log(x+1)$, which is concave: nothing is hidden in it. The non-monotonicity is entirely the budget's, and the threshold is sharp, τ against R. A taper gentler than the gross return leaves the household's saving monotone in assets no matter how badly the test distorts the value function; a taper steeper than the gross return, or a cliff, makes a richer household poorer in the only sense that matters for today's decision.

5.3 Which policy, and in which variable

The two non-monotonicity results are about different objects, and it is worth separating them in the household's own state space, assets a and the Markov income shock z.

Next period's assets are monotone. For each z, the optimal a' is nondecreasing in a whatever the continuation value, with no concavity assumed of it:

Theorem 5.4 (a' rises with a, at each z)**.** *Let $a < a'$ both lie in the asset region. Then for every continuation v and every z, $g_v(a, z) \le g_v(a', z)$.*
Formal statement: *`IncomeFluctuation.policyOf_mono_of_lt`.*

This is Theorem 5.1 read in the state space: cash on hand rises with assets, so optimal saving does. The repository already had this statement (`policyOf_mono`) but with concave slices of the continuation as a hypothesis, which is exactly what a means test destroys; the Topkis argument removes the hypothesis, at the cost of requiring the two asset levels to be distinct. The continuation cancels between the two optimality inequalities and only strict concavity of period utility survives.

Consumption is not monotone, and the contrast is sharp rather than an artefact of proof technique. Consumption is $m(a) - g_v(a, z)$, and both terms rise with a; when the continuation is non-concave the second can rise faster, which is exactly Theorem 5.2. So monotone consumption genuinely requires concavity of the continuation, while monotone saving does not.

Two things follow for the means test. Where the pension enters only next period's problem, so that today's cash on hand is unaffected, Theorem 5.4 applies verbatim: next period's assets still rise with this period's, and only consumption misbehaves. Where the test is assessed on the assets currently held, the hypothesis of Theorem 5.4 is the one that fails: cash on hand is no longer increasing in a, and with it the policy. That is the whole of Theorem 5.3, and the dividing line is τ against R.

5.4 At realistic parameters

The three theorems predict exactly which of the two channels operates for which schedule, and a calibrated retiree confirms it. We solve a deterministic retiree of 25 years by backward induction with global search over a grid of 6001 asset levels — the value function is not concave, so a first-order condition cannot be used — at $r = 4\%$, $\beta = 0.96$, relative risk aversion 3, a full pension of 0.40 of mean earnings and a free area of 3 mean-earnings units (`numerics/meanstest.m`). The taper of 0.078 per unit of assets is Australia's asset test, three dollars a fortnight per thousand dollars.

Pension schedule	Cash on hand	Saving	Consumption
Lump sum	increasing	increasing	increasing
Taper 0.078 (Australia)	increasing	increasing	falls by 0.025–0.046
Taper 0.50	increasing	increasing	falls by 0.13–0.24
Taper 1.20 (above R)	falls	falls by 0.004	falls by 0.55–0.70
Cliff	falls by 0.40	falls by 0.32–0.37	falls by 0.55–0.70

Every entry is what the theorems require. Saving falls with assets in exactly the two rows where cash on hand does, which are exactly the two rows whose taper exceeds the gross return, as Theorem 5.3 says; in the other three it rises, as Theorem 5.1 says it must. Consumption falls in every row with a test of any kind, as Theorem 5.2 says it can, and the drop sits exactly at the free area and grows with age, from 0.025 of mean earnings at the start of retirement to 0.046 twenty years in under the Australian taper. The mechanism is the one the witness isolates: just above the free area the effective return to saving is only $R - \tau$ until the pension is exhausted, so the household jumps its saving toward the far side of the taper range and cuts consumption to pay for it.

The effect is not confined to retirees. A worker in the last year before retirement, facing the Australian taper for the whole of retirement, has a monotone saving policy and a consumption policy that falls by 0.025 of mean earnings at the same threshold: the non-concavity is inherited through the continuation value, and it is again consumption that carries it.

6 Stationary equilibrium

Every cohort is born with no assets and an earnings state drawn from a fixed law ν, lives J periods, and is replaced; cohorts have equal weight. The distribution of the youngest cohort is therefore the same at every interest rate, and the distribution of every older cohort is that law pushed forward by the policies of the ages in between. This is the structural advantage of the life cycle over the infinite horizon, and it removes most of the machinery the stationary case needed.

Two of the consumption bounds this section uses are developed in Section 7, where they arise in the hunt for the critical risk aversion: the constraint-incidence bound of Theorem 7.9 and the asset-dependent saving floor of Section 7.16. Nothing here depends on the verdict of that section.

6.1 The aggregate is a finite composition

Because assets at birth are degenerate at zero, no measure theory is required. Let

$$(\mathcal{S}_k h)(a,z) = \sum_{z'} \pi(z,z')\, h\big(g_k(a,z),z'\big)$$

be the expectation of h next period for a household with k periods still to come (`stageStep`). The assets a household holds over the rest of its life, summed across ages and seen from its current state, satisfy

$$A_0(a,z) = a, \qquad A_{k+1}(a,z) = a + (\mathcal{S}_{k+1}A_k)(a,z)$$

(`cohortAssets`), and aggregate capital per head is $K = \frac{1}{J}\sum_z \nu(z)\, A_{J-1}(0,z)$ (`olgCapital`). There is no pushforward measure, no Feller property, no stationary distribution, no Doeblin condition and no uniqueness of an invariant law: the equilibrium object is a finite sum of finite sums, and the recursion above is its definition.

6.2 The chain

Theorem 6.1 (Light's Theorem 2 for the life cycle)**.** *Suppose two economies share the earnings chain and their policies are ordered at every age, $g_k^{(1)} \le g_k^{(2)}$ on the asset region. Then $A_k^{(1)} \le A_k^{(2)}$ for every k, and aggregate capital is ordered, $K^{(1)} \le K^{(2)}$.*
Formal statement: *`IncomeFluctuation.cohortAssets_le, IncomeFluctuation.olgCapital_le_of_stagePolicy_le`.*

The proof is an induction on age with two ingredients, and both are already available.

The class of functions increasing in assets is preserved by $\mathcal{S}_k$ (`monoRegion_stageStep`), because next period's assets rise with this period's at each earnings state and for any continuation, which is Theorem 5.4. And ordered policies keep ordered functions ordered (`stageStep_le_stageStep`): the higher-rate economy evaluates a larger function at a larger asset level. Note what is *not* needed. The infinite-horizon version of this step required monotone transitions, a coupling argument and first-order stochastic dominance of the stationary distributions; here the whole statement is an inequality between two functions on the same finite state space, because the distribution the two economies start from is literally the same.

6.3 Uniqueness

Theorem 6.2 (Single crossing)**.** *A capital supply that never falls with the rate meets a strictly falling demand at most once.*
Formal statement: *IncomeFluctuation.olgEquilibriumRate_unique.*

With Theorem 6.1 this reduces uniqueness of the stationary equilibrium to Light's Theorem 1, that saving rises with the rate at every age, which is the subject of Section 7: it holds at Aiyagari's calibration for relative risk aversion up to about 3.4, so the chain closes at his $\mu = 3$ and not at his $\mu = 5$.

6.4 Continuity, and existence

The last input is continuity of capital supply in the rate. In the infinite horizon this was a substantial development, because the stationary distribution itself had to be shown to move continuously. Here there is no stationary distribution: capital supply is a finite composition of finitely many policies, so all that is needed is that each policy moves continuously with the rate, and that continuity composes.

We take the route of the companion write-up and put the interest rate into the state, so that Berge's theorem gives continuity in assets and the rate jointly rather than continuity in a parameter. That device was used there at the fixed point, but everything it rests on — that the feasible sets and the objectives agree on the slice — holds for an arbitrary continuation. So it works stage by stage, and here it is easier, because the slice of the augmented Bellman iterate is the iterate of the sliced programme by a plain induction, with no fixed point to identify (`sliceAt_bellman`, `sliceAt_augStageValue`). The augmented argmax at each stage is then a singleton, since the sliced one is by Carroll–Kimball concavity of the iterates (`argmax_aug_stage_eq_singleton`), so the stage policy is jointly continuous (`continuousOn_augStagePolicy`) and agrees with the sliced one (`augStagePolicy_eq`). Running the cohort recursion in the augmented state (`augCohortAssets`) gives a composition of continuous maps that reads, at a rate in range, as capital supply at that rate (`augCohortAssets_eq`, `olgSupply_eq`).

Theorem 6.3 (Capital supply is continuous)**.** $r \mapsto K(r)$ *is continuous on* $[r_{lo}, r_{hi}]$.
Formal statement: *IncomeFluctuation.continuousOn_olgSupply.*

Theorem 6.4 (Existence)**.** *If capital supply lies below demand at the bottom of the interval and above it at the top, and demand is continuous, then the two meet: there is a rate in* $[r_{lo}, r_{hi}]$ *at which the capital the cohorts accumulate equals the capital the firm demands.*
Formal statement: *IncomeFluctuation.exists_olgEquilibrium*.

Both hypotheses of Theorem 6.4 are about the solved model. The rest of this section replaces them with inequalities that never mention the household's policy or value function.

6.5 Where the household can be

Proposition 6.9 inherits from Theorem 3.1 the hypothesis that the artefactual cap is slack, and the convenient way to state that is slack everywhere in $[0, \bar{a}]$. That hypothesis is not innocent, and it fails exactly where the floor is wanted.

The one-step bound gives $g_k(a, z) \le (1 - \kappa_k)(Ra + y_{\max})$, so $[0, \bar{a}]$ is carried into itself only if $(1 - \kappa_k)(R\bar{a} + y_{\max}) \le \bar{a}$, which needs $(1 - \kappa_k)R < 1$. As k grows κ_k falls to $1 - \text{þ}/R$, so the requirement is $\text{þ} < 1$, that is $\beta R < 1$: the household must be impatient. Above the rate of time preference no cap is forward invariant at any level, because a patient household sitting at the cap wants to save past it. The cap then binds, the Euler inequality under it is unavailable, and the slackness hypothesis is false rather than merely unverified. A theorem carrying it says nothing there.

This matters because a stationary life-cycle equilibrium sits above the rate of time preference. In the calibration of Section 6.10 with $\lambda = 1/\beta - 1 = 4.17\%$, capital supply crosses demand near 5%, and the rate at which the *provable* floor crosses demand is near 9%. Both are in the region where no cap is slack everywhere.

A cohort that begins life with nothing never visits the top of $[0, \bar{a}]$. The development therefore carries a family of regions, one per stage, that the stage policies map down through, and asks for slackness only on those.

Definition 6.5 (Reachable family)**.** A reachable family assigns to each stage k a set of asset levels such that every set sits inside $[0, \bar{a}]$, the stage-$(k+1)$ policy carries the stage-$(k+1)$ set into the stage-k set, and $g_k < \bar{a}$ on every set.
Formal statement: `IncomeFluctuation.StageRegions`.

The whole interval $[0, \bar{a}]$ is a reachable family exactly when the cap is slack on it (`allRegions`), so nothing is lost: Theorem 3.1 and Proposition 6.9 are the constant-family instances of statements that hold on any reachable family (`stageConsumption_le_stageHumanWealth_on`, `le_stagePolicy_on`).

What a cohort visits is bounded by feasibility alone. Set $\rho_0 = 0$ and $\rho_{j+1} = R\rho_j + y_{\max}$: a household that cannot carry forward more than its cash on hand holds at most ρ_j after j periods of life, whatever the earnings draws.

Proposition 6.6 (A cohort's own regions)**.** *If* $\rho_{K+1} < \bar{a}$ *then the sets* $[0, \rho_{K-k}]$*, empty for* $k > K$*, form a reachable family for a cohort of length* $K + 1$ *that starts life with nothing.*
Formal statement: *IncomeFluctuation.reachRegions*.

The bound is deliberately crude, using none of the damping that optimal saving supplies, and that is what keeps it free of the sandwich it is used to prove. Its cost is a larger cap. At $r = 10\%$ over sixty periods ρ_{K+1} is about 9,044 where the optimal path reaches about 82. Since the cap is an artefact of the state space rather than an economic object, a cap of ten thousand is as acceptable as a cap of one hundred, and unlike slackness everywhere it is a condition that can actually be met. The sandwich and the floor then hold along a cohort's path with no hypothesis on the cap beyond $\rho_{K+1} < \bar{a}$ (`stageConsumption_le_stageHumanWealth_reach`, `le_stagePolicy_reach`).

At the bottom of the rate interval the same crude bound does the other half of the work on its own. When $R < 1$ the sequence ρ_j converges to $y_{\max}/(1-R)$, so cohort assets are bounded without any appeal to optimality: at $r = -7\%$ the bound is 42.1 against a capital demand of 56.3.

6.6 The propensity in closed form

Both bounds below are built from the finite-horizon propensity κ_k and from human wealth, and both need to know not just where κ_k sits but how fast it settles.

Proposition 6.7 (The finite-horizon propensity)**.** *Write $q = \beta/R$. Then $\kappa_k \sum_{i \le k} q^i = 1$, and hence $\kappa_k(1 - q^{k+1}) = 1 - q$: the propensity exceeds its infinite-horizon limit $1 - q$ by exactly $\kappa_k q^{k+1}$.*
Formal statement: *`IncomeFluctuation.stageMPC_mul_mpcSum`*.

`, IncomeFluctuation.stageMPC_mul_one_sub_pow`

The defining recursion $\kappa_{k+1} = \kappa_k R/(\beta + \kappa_k R)$ is exactly $1/\kappa_{k+1} = 1 + q/\kappa_k$, so the reciprocal accumulates the geometric series and nothing else. The same sum tames the slope that both bounds share. Writing $S_k = \sum_{i \le k} q^i$, the recursion $G_0 = 1$, $G_{k+1} = 1 + G_k(1-\kappa_{k+1})R$ has an age-dependent coefficient, but $W_k = G_k S_k$ obeys $W_{k+1} = S_{k+1} + \beta W_k$, which does not. Then $S_{k+1} \ge 1$ and $(1-q)S_k \le 1$ give the bound that matters.

Proposition 6.8 (A slope that grows)**.** $G_k \;\ge\; (1-q)\sum_{j \le k} \beta^j$.
Formal statement: *`IncomeFluctuation.mul_geomSum_le_cohortSlope`*.

Where the crude $G_k \ge 1$ does not grow at all, this grows geometrically, and Section 6.11 reports what that is worth.

6.7 A floor on capital

Existence needs capital supply above demand at the top of the rate interval. In the infinite horizon the only floor available came from an ergodic identity that loses a factor $\|u'\|_\infty^2/\operatorname{Var}$. The life cycle needs nothing of the kind: the upper sandwich of Theorem 3.1 bounds consumption at every age, and what is not consumed is saved.

Proposition 6.9 (Affine floor on saving)**.** *At every age and every state in the region,*

$$g_k(a, z) \;\ge\; (1 - \kappa_k)(y_z + Ra) - \kappa_k H_k(z).$$

Formal statement: *IncomeFluctuation.le_stagePolicy_on.*

, IncomeFluctuation.le_stagePolicy

Iterating it is not a matter of tracking a distribution over assets. An affine lower bound on a cohort's accumulated assets suffices, $F_k(z) + G_k a \le A_k(a, z)$, with the slope G of Section 6.6 common to all earnings states and an intercept carrying one value per state:

$$F_0 = 0, \qquad F_{k+1}(z) = \textstyle\sum_{z'} \pi(z, z') F_k(z') + G_k\big[(1 - \kappa_{k+1}) y_z - \kappa_{k+1} H_{k+1}(z)\big].$$

Proposition 6.10 (Affine floor on accumulated assets)**.** *On any reachable family, $F_k(z) + G_k a \le A_k(a, z)$; hence capital per head is at least $\sum_z \nu_z F_K(z)/(K+1)$.*
Formal statement: *IncomeFluctuation.cohortFloor_add_mul_le_cohortAssets.*

, IncomeFluctuation.le_olgCapital

The state dependence of F is the whole content, not a refinement. Reading Proposition 6.9 at the worst earnings state instead, which is the only way to get a bound uniform in z, gives a floor that is identically zero at every rate in the calibration of Section 6.10, against 4.75 at $r = 10\%$ for the state-dependent one.

Averaging against weights the chain leaves alone collapses the vector recursion to a scalar one, and the sign of each increment then has an exact answer.

Theorem 6.11 (The floor rises exactly when the household is patient)**.** *Let $\bar{y}$ and $\bar{H}_k$ be the averages of earnings and of human wealth under weights ν satisfying $\nu\pi = \nu$. If $\text{þ} \ge 1$, equivalently $\beta R \ge 1$, then $\kappa_k \bar{H}_k \le (1 - \kappa_k)\bar{y}$ at every age, so the aggregate floor never falls and is never negative.*
Formal statement: *IncomeFluctuation.stageMPC_mul_sum_stageHumanWealth_le.*

, IncomeFluctuation.sum_mul_cohortFloor_mono

The increment of the aggregate floor is $G_k \kappa_{k+1} \bar{y}$ times the slack in that inequality, and the inequality reverses below $\text{þ} = 1$. So the provable floor is flat under the rate of time preference and climbs above it, which is why the upper endpoint of the rate interval has to be taken above $\lambda = 1/\beta - 1$. The proof is one induction: writing $v_k = (1 - \kappa_k)/\kappa_k$ and $u_k = \bar{H}_k/\bar{y}$, both satisfy first-order recursions, $v_{k+1} = (\text{þ}/R)(1 + v_k)$ and $u_{k+1} = (1 + u_k)/R$, from the same starting point, and $\text{þ} \ge 1$ keeps the first above the second.

One thing the affine form cannot do is refuse to claim negative assets. Its intercept goes badly negative: at the calibration, 160 of 413 entries are below zero and the worst is -62, with F_K running from -58.8 at the lowest earnings to $+685$ at the highest. A cohort's accumulated assets cannot be negative, so all of that is slack the aggregate inherits. Reading the floor as a function of assets rather than as an affine form fixes it, because the saving floor can then be clamped at zero where it belongs:

$$\Lambda_0(z, a) = a, \qquad \Lambda_{k+1}(z, a) = a + \textstyle\sum_{z'} \pi(z, z')\, \Lambda_k\big(z', \Phi_{k+1}(z, a)\big),$$

with Φ the clamped saving floor of Section 7.16. The proof needs only that Λ_k is increasing in assets, which the clamp preserves.

Proposition 6.12 (The clamped floor)**.** $\Lambda_k(z,a) \le A_k(a,z)$ *on any reachable family, so capital per head is at least* $\sum_z \nu_z \Lambda_K(z,0)/(K+1)$*, which is at least what Proposition 6.10 gives.*

Formal statement: *IncomeFluctuation.cohortFloorAt_le_cohortAssets.*

, IncomeFluctuation.le_olgCapital_at

6.8 A ceiling on capital

Existence also needs capital supply below demand at the bottom of the interval, and there three bounds are available, each asking for more and giving more.

The cheapest uses the budget and nothing else. A household cannot carry forward more than its cash on hand, so an asset level B with $RB + y_{\max} \le B$ is never left once entered.

Proposition 6.13 (A forward-invariant level caps supply)**.** *If* $0 \le B \le \bar{a}$ *and* $RB + y_{\max} \le B$*, then capital per head is at most* B*.*

Formal statement: *IncomeFluctuation.olgCapital_le_of_invariant.*

The second says the household will not want to carry that much. Consuming at least $\kappa_k(m + H_k(z))$ by the constraint-incidence bound of Theorem 7.9, it saves at most $(1 - \kappa_k)(y_z + Ra) - \kappa_k H_k(z)$.

Proposition 6.14 (The incidence ceiling)**.** *If* $B \le \bar{a}$ *and, at every age and earnings state,* $(1-\kappa_j)(y_w + RB) - \kappa_j H_j(w) \le B$*, then capital per head is at most* B*.*

Formal statement: *IncomeFluctuation.olgCapital_le_of_incidence.*

Both bound *every* age's assets by one level, and so pay $(K+1)B$ for a cohort that holds nothing when it is born, nothing when it dies, and anything like B only in between. The floor never made that mistake, and the third ceiling does not either: it is the same affine object, built from the same slope G, with the saving ceiling in place of the saving floor,

$$\bar{F}_0 = 0, \qquad \bar{F}_{k+1}(z) = \textstyle\sum_{z'} \pi(z,z') \bar{F}_k(z') + G_k\big[(1-\kappa_{k+1})y_z - \kappa_{k+1} H_{k+1}(z)\big],$$

with H here the capped, risk-adjusted human wealth of Theorem 7.9.

Proposition 6.15 (The affine ceiling)**.** $A_k(a,z) \le \bar{F}_k(z) + G_k a$*, and hence capital per head is at most* $\sum_z \nu_z \bar{F}_K(z)/(K+1)$*.*

Formal statement: *IncomeFluctuation.cohortAssets_le_cohortCeil.*

, IncomeFluctuation.olgCapital_le_cohortCeil

Replacing a worst case over ages by an average over them is worth a factor of fourteen, and it keeps working above the rate of time preference, where no forward-invariant level exists at all.

	$r = -7\%$	$r = -4\%$	$r = 0\%$	$r = \lambda$	$r = 5\%$
Capital demand	56.25	14.06	7.03	4.62	4.33
Proposition 6.13, the budget	42.52	74.41	—	—	—
Proposition 6.14, incidence	22.17	35.11	—	—	—
Proposition 6.15, affine	1.54	2.28	3.86	5.98	6.47
True capital supply	0.64	1.03	1.98	3.78	4.25

6.9 Existence from primitives

The two ends can now be put together, and the statement has taken four forms as the bounds improved. All four are in the development; the last is the one to read.

Theorem 6.16 (Existence from primitives)**.** *Fix a rate interval and continuous demand D. Suppose there is a B with $0 \le B \le \bar{a}$, $(1 + r_{lo})B + y_{\max} \le B$ and $B \le D(r_{lo})$; and suppose $\rho_{K+1} < \bar{a}$ and $D(r_{hi}) \le \sum_z \nu_z F_K(z)/(K+1)$ at r_{hi}. Then there is a rate in the interval at which the capital the cohorts accumulate equals the capital the firm demands.*
Formal statement: *IncomeFluctuation.exists_olgEquilibrium_rate.*

, IncomeFluctuation.exists_olgEquilibrium_of_bounds

Nothing in those hypotheses refers to the household's decisions. They are inequalities among the discount factor, the interest rate, the earnings process, the horizon and the demand curve, with ρ, F and G explicit finite recursions in those primitives. Three refinements follow. Proposition 6.8 turns the floor into a closed form with no recursion left to run (`IncomeFluctuation.exists_olgEquilibrium_of_geomBounds`, Theorem 6.19 below); and Proposition 6.14 replaces the budget ceiling (`IncomeFluctuation.exists_olgEquilibrium_of_sharpBounds`). The affine ceiling replaces it again, and at that point no auxiliary B and no invariance condition survive.

Theorem 6.17 (Existence, both ends read the same way)**.** *With the ceiling at r_{lo} and the floor at r_{hi} both taken as weighted averages of affine bounds on a cohort's accumulated assets, a stationary equilibrium exists.*
Formal statement: *IncomeFluctuation.exists_olgEquilibrium_affine.*

Reading the floor at the assets held as well gives the tightest form.

Theorem 6.18 (Existence with both ends read at the assets held)**.** *With the affine ceiling at r_{lo} and the clamped floor at r_{hi}, a stationary equilibrium exists.*
Formal statement: *IncomeFluctuation.exists_olgEquilibrium_clamped.*

The two boundary conditions are then the same object built twice, once from the saving ceiling and once from the saving floor, sharing a slope. Each half asks for what it can have: at the bottom the artefactual cap is slack, which holds below the rate of time preference; at the top it cannot be, and the reachable family of Section 6.5 stands in. So the low endpoint must sit below λ and the high endpoint above it, and Section 6.11 shows a bracket that does.

For completeness, the closed-form variant.

Theorem 6.19 (Existence in closed form)**.** *Under the conditions of Theorem 6.16, with the floor hypothesis replaced by the closed-form expression of Proposition 6.8 and with $1 \le \underline{p} \le R$ at r_{hi}, a stationary equilibrium exists.*
Formal statement: *IncomeFluctuation.exists_olgEquilibrium_of_geomBounds.*

6.10 What the numbers say

The calibration is the one used throughout: $J = 60$, $\beta = 0.96$, relative risk aversion 3, the Tauchen discretisation of Aiyagari's earnings process with $\rho = 0.9$ and $\sigma = 0.4$, and a

Cobb–Douglas technology with $\alpha = 9/25$ and $\delta = 2/25$, so capital demand is $\alpha/((1-\alpha)(r+\delta))$ and $\lambda = 4.17\%$. Capital supply is the average asset holding across the sixty cohorts, each born with nothing. The last row is not the solved supply but the floor of Proposition 6.10 (`numerics/olgexist.m`).

r	-4%	0%	λ	6%	8%	10%	15%
Capital demand	14.06	7.03	4.62	4.02	3.52	3.13	2.45
Capital supply	1.03	1.98	3.78	4.85	6.18	7.69	12.49
Provable floor	0.00	0.00	0.00	1.52	3.13	4.75	9.40

Supply crosses demand at 5.09%, above the rate of time preference, which is what Section 6.5 has to accommodate; and the floor is flat below λ and climbs above it, which is Theorem 6.11. The optimal path reaches assets of 82 at $r = 10\%$, against the crude reachable bound of 9,033; the constants are ordinary, unlike the stationary case, where the corresponding instantiation put r_{hi} within 10^{-27} of λ.

6.11 The theorem checked, hypothesis by hypothesis

Take $r_{lo} = 2\%$, $r_{hi} = 8\%$ and an artefactual cap of 10^4. Every hypothesis of Theorem 6.18 holds (`numerics/clampcheck.m`): the cap is forward invariant at r_{lo} with margin 134; the ceiling is 4.81 against a demand of 5.63; a lifetime of feasible saving reaches 3,730 against the cap; and the floor is 3.80 against a demand of 3.52. The bracket contains the true equilibrium at 5.09%.

Sweeping the endpoints shows what each bound is worth. The low endpoint may be taken as high as

Ceiling used	highest r_{lo}	Floor used	lowest r_{hi}
Proposition 6.13, the budget	-6.73%	closed form, crude $G_k \geq 1$	610%
Proposition 6.14, incidence	-5.85%	closed form, Proposition 6.8	18.2%
Proposition 6.15, affine	$+2.78\%$	Proposition 6.10, affine	8.38%
		Proposition 6.12, clamped	7.67%

So the tightest bracket the development gives is [2.78%, 7.67%], and the absolute limit is 5.09% from both sides, since no bound on capital supply can cross demand before supply does. The closed-form floor is reported because it needs no recursion at all; at a calibrated rate it is much the weaker, giving 0.48 at $r = 10\%$ against a true floor of 4.75.

Neither bracket is tight, and the same fact stops them at both ends. Appendix A gives it: the obstruction at the top and the obstruction at the bottom are one obstruction, that a power mean with negative exponent is superadditive rather than 1-Lipschitz.

7 Theorem 1 in the life cycle: how far it goes

The programme of Section 11 rests on Light's Theorem 1, that saving rises with the interest rate at every state. In the infinite horizon this is open above logarithmic utility. The life cycle

was the reason for hope: a finite backward induction from an explicit terminal condition, with the per-step loss compounding only J times instead of to a fixed point. This section reports what the numbers say, at $\beta = 0.96$, a seven-state Tauchen chain with $\rho = 0.9$ and $\sigma = 0.4$, a zero borrowing limit, $J = 60$, and a flat earnings profile (`numerics/lc_check.m`, `lc_robust.m`, `lc_induct.m`, `lc_verify.m`, `lc_bound.m`, `lc_crit.m`).

7.1 The statement is its own terminal condition

The budget constraint makes two statements identical. Since $a' = Ra + y(z) - c$ at either rate,

$$g_k(a, z; r_2) \;\geq\; g_k(a, z; r_1) \qquad \Longleftrightarrow \qquad c_k(a, z; r_2) - c_k(a, z; r_1) \;\leq\; (r_2 - r_1)\, a,$$

so Theorem 1 *is* the affine bound on the rate response of consumption with slope one and zero intercept. That bound holds with equality in the last period of life, where $c_0 = y(z) + Ra$. The question is therefore whether the terminal condition propagates backward, and it is a question about one inequality rather than about a pair of constants to be chosen.

7.2 It propagates, up to a critical risk aversion

At $\mu = 1$ and $\mu = 3$ it propagates exactly. Over all sixty stages and the whole wealth region reachable from zero wealth at birth, the largest violation of $g_k^{(1)} \leq g_k^{(2)}$ is zero, and the consumption bound holds to five parts in 10^{14}; refining the asset grid from 2000 to 16000 points changes nothing. At $\mu = 5$ it fails. The failure is a genuine feature of the household's problem, not of the solution method: the violation is identical at every grid size, and endogenous-grid and value-function iteration with a pure grid search agree on it to the resolution of the grid. Saving at zero wealth in the highest income state, in the first period of a sixty-period life, falls as the rate rises:

r	2%	3%	4%	5%	6%
$g(0, z_{\max})$, endogenous grid	1.7783	1.7638	1.7452	1.7234	1.6993
$g(0, z_{\max})$, value iteration	1.7806	1.7606	1.7406	1.7206	1.7006

Between the two lies a critical degree of risk aversion, and it is sharp. Bisecting on μ for the largest violation over all stages gives:

Horizon J	$r = 2\%$	$r = 4\%$	$r = \lambda = 4.17\%$
20	3.79	3.53	3.51
40	3.77	3.39	3.36
60	3.77	3.39	3.36

Theorem 1 holds throughout the life cycle below the critical value and fails above it. The value is stable in the horizon beyond about forty periods, and falls as the rate rises towards the rate of time preference. Aiyagari's $\mu = 3$ is inside it with little room, at $\mu = 3.39$ against

3; his $\mu = 5$ is well outside. So the life cycle buys one of the two open cases and not the other, and it buys $\mu = 3$ only at rates below the rate of time preference, which is where the equilibrium lies.

7.3 The reduction, formalised

The natural backward induction on the response of consumption to the rate does not close; its step multiplier exceeds one, and Appendix A gives the reason and the numbers. What is proved in Lean is a different form, the one the life cycle makes natural. Because the budget constraint makes the two statements identical (`stagePolicy_le_iff_consumption_le`), the terminal stage needs no hypothesis at all: in the last period both households consume their cash on hand and save nothing (`stagePolicy_mono_zero`). Corners need none either: if the lower-rate household saves nothing there is nothing to violate. What the step needs is one inequality per age and state, which we call the *age-indexed slack condition*: writing b for the lower-rate household's saving at that state and $C'(z')$ for its consumption next period,

$$R_1 \sum_{z'} \pi(z,z') C'(z')^{-\mu} \;\le\; R_2 \sum_{z'} \pi(z,z') \big(C'(z') + \Delta r\, b\big)^{-\mu} \Big(1 + \frac{\Delta r\, a}{c}\Big)^{\mu} \tag{1}$$

(`StageSlack`).

Theorem 7.1 (The reduction)**.** *If* (1) *holds at every age and every state in the region, then saving rises with the rate at every age, and aggregate capital rises with the rate.*
Formal statement: *`IncomeFluctuation.stagePolicy_mono_step`, `IncomeFluctuation.stagePolicy_mono_of_stageSlack`, `IncomeFluctuation.olgCapital_le_of_stageSlack`.*

With Theorem 6.2 this closes the chain: (1) at every age gives a unique equilibrium rate. The question is therefore whether (1) is true, and the answer is the one reported above, now stated sharply. Condition (1) is exactly the margin tabulated in the previous subsection. At $\mu = 3$ it holds for the last three or four periods of life, with margins $+0.68$, $+0.42$ and $+0.18$ per unit of the rate gap, and fails from about the fifth-to-last, at -0.23, deteriorating to -2.86 at birth. At $\mu = 1$ it holds for most of the life and still fails for the youngest ages, at -0.12 at birth.

That last observation is the important one. The condition fails even at logarithmic utility, where Theorem 1 is certainly true. So the failure is not a symptom of high risk aversion: the pointwise Euler comparison is simply the wrong instrument in a life cycle, and no sharpening of it that we could find repairs the young ages. Keeping the term the step discards, the marginal propensity to consume times the size of the hypothetical violation, only bounds the violation by a multiple of Δr, and both refinements we tried — an upper bound on the propensity to consume from concavity and the sandwich, and a self-improving bootstrap on the size of the violation — converge to a positive bound rather than to zero.

So $\mu = 3$ is not delivered by this route, and $\mu = 1$ is not either, although the companion write-up proves the latter by the rescaling argument, which is an induction on Bellman

iterates and so carries over to the life cycle unchanged. What the reduction does deliver is the end of life, and the exact inequality that any proof must beat.

So the state of the programme is this. Theorem 1 is true in the life cycle for $\mu = 3$, and the statement to prove has the cleanest possible form, its own terminal condition propagated; the reduction to one inequality per age is proved; and that inequality is false for the young, at every degree of risk aversion. What is missing is an argument at zero wealth that does not go through a pointwise comparison of the two Euler equations. The same zero-wealth state is where the infinite-horizon slack condition was tightest, which suggests the two problems share a single missing ingredient rather than two.

7.4 It is not precautionary saving

A household at zero wealth in the highest earnings state saves heavily, and the obvious diagnosis is precautionary: it is insuring against the fall in earnings it expects. If that were the mechanism, precautionary-saving theory would be the place to look for the missing argument. It is not the mechanism. The same reversal happens in the *certainty-equivalent* household, which faces the conditional mean earnings path and no risk at all, and there everything is in closed form (`numerics/precaut2.m`, `durcheck.m`).

d(saving)/dr at zero wealth	$\mu = 1$	$\mu = 3$	$\mu = 5$	$\mu = 8$
Certainty equivalent (mean path, no risk)	+21.3	+2.81	−0.85	−2.91
With risk	+17.1	+0.92	−2.05	−3.43

The sign flips between $\mu = 3$ and $\mu = 5$ in both columns. Bisecting, the certainty-equivalent threshold is $\mu = 4.33$ against $\mu = 3.39$ with risk: risk moves the threshold by about one unit of risk aversion and does not create the effect.

Without risk the mechanism is exact. A household with J periods, no borrowing constraint and CRRA utility has $c_{t+1} = (\beta R)^{1/\mu} c_t$ and a lifetime budget, so

$$c_0 = \frac{W(R)}{D(R)}, \qquad W(R) = \sum_t y_t R^{-t}, \qquad D(R) = \sum_t \beta^{t/\mu} R^{t/\mu - t}$$

(`consumption_mul_pathPrice`), and saving at zero wealth rises with the rate exactly when $W(R_2)D(R_1) \le W(R_1)D(R_2)$ (`consumption_antitone_iff`). Both terms fall with the rate, W because future earnings are discounted harder and D because the optimal path gets cheaper, and the comparison is between the two elasticities. Differentiating,

$$\frac{\mathrm{d}\log c_0}{\mathrm{d}\log R} = -T_y + \Big(1 - \frac{1}{\mu}\Big) T_c,$$

with T_y and T_c the Macaulay durations of the two streams, so c_0 falls, and saving rises, exactly when $T_y > (1 - 1/\mu)T_c$. The predicted elasticity matches the exact one to three decimals at every μ we checked. At Aiyagari's numbers $T_y = 14.31$ and $T_c \approx 18.6$, nearly independent of μ, so the condition is $1 - 1/\mu < 0.769$, that is $\mu < 4.33$.

The economics is then plain, and has nothing to do with risk. A higher interest rate discounts future earnings more heavily, which makes the household poorer and pushes consumption down; it also makes a rising consumption path cheaper, which pushes consumption today down only to the extent that the household is willing to tilt its path. A more risk averse household will not tilt, so the second force weakens as μ grows while the first does not, and past $\mu \approx 4$ the household responds to a higher rate by consuming more today. The income stream simply has to be longer than the consumption weights, discounted by the strength of the tilt.

Theorem 7.2 (No risk, and risk aversion at most one)**.** *If $\mu \le 1$ the comparison holds term by term, whatever the earnings profile: the certainty-equivalent household always saves more at a higher rate.*
Formal statement: *`CertaintyEquivalent.humanWealth_mul_pathPrice_le_of_le_one`, `CertaintyEquivalent.consumption_antitone_of_le_one`.*

For $\mu > 1$ it is the duration comparison, which is a finite computation in the primitives. So a precautionary argument near zero wealth would be aimed at the wrong target: what has to be controlled is the elasticity of human wealth against the elasticity of the price of the consumption path, and risk enters only as a correction to it.

7.5 The criterion, and the risk correction

The certainty-equivalent criterion is cleanest when written in the intertemporal elasticity of substitution rather than in risk aversion. Since it is linear in $1/\mu$,

$$T_y > \Big(1 - \frac{1}{\mu}\Big) T_c \qquad \Longleftrightarrow \qquad \frac{1}{\mu} \;>\; 1 - \frac{T_y}{T_c},$$

so the elasticity of substitution must exceed one minus the ratio of the two durations. At Aiyagari's numbers that right-hand side is 0.2306, giving $\mu < 4.33$ as before, and the identity holds to four decimals against the bisected threshold.

With risk the threshold moves down, and the move is what has to be bounded. Write

$$\Delta \;:=\; \frac{1}{\mu^{\text{risky}}} - \frac{1}{\mu^{\text{CE}}}$$

for the gap in elasticity units and $s = \sigma\sqrt{1-\rho^2}$ for the conditional standard deviation of log earnings, the dispersion the household actually faces one period ahead. Measuring both thresholds at the same state over fifteen calibrations, $\rho \in \{0, 0.6, 0.9\}$ and σ from 0.025 to 0.4 (`numerics/gapscale.m`, `gapfit.m`):

$\rho = 0.9$, σ	0.025	0.05	0.1	0.2	0.4	fitted exponent
Δ	0.0034	0.0070	0.0144	0.0300	0.0612	
Δ/s	0.315	0.320	0.330	0.344	0.351	1.04

The gap is first order in the conditional standard deviation, not second order: fitting $\Delta \sim s^b$ gives $b = 1.04$ at $\rho = 0.9$, 0.89 at $\rho = 0.6$ and 0.93 at $\rho = 0$, and the ratio Δ/s

stays between 0.196 and 0.351 across every calibration we ran. That first-order behaviour is itself informative: a precautionary effect on saving would be second order in the dispersion, whereas the probability that the borrowing constraint binds in a low-earnings state next period moves at first order, and that is our reading of where the correction comes from.

Taking the largest ratio observed as the bound, $\Delta \le 0.36\, s$, gives a criterion entirely in the primitives:

$$\frac{1}{\mu} \;\ge\; 1 - \frac{T_y}{T_c} \;+\; 0.36\,\sigma\sqrt{1-\rho^2}. \tag{2}$$

At Aiyagari's calibration the right-hand side is $0.2306 + 0.0628 = 0.2934$, that is $\mu \le 3.41$, against the 3.39 that bisection on the full life-cycle model gives. The bound therefore reproduces the measured threshold to within two per cent, and it says the same thing the earlier sections said: the calibration he used for $\mu = 3$ is inside, and $\mu = 5$ is not.

7.6 The certainty-equivalent half, for every risk aversion

The first inequality of (2) was a theorem only for $\mu \le 1$, where the exponent $\theta = 1 - 1/\mu$ is nonpositive and every factor moves the same way. It is now a theorem for every μ. Writing $\lambda = R_1/R_2 < 1$, the criterion $W(R_2)D(R_1) \le W(R_1)D(R_2)$ compares how the same weights respond to λ^t against $(\lambda^t)^\theta$. Each λ^t lies in $[\lambda^{J-1}, 1]$, and on that interval the concave $x \mapsto x^\theta$ lies above its chord, which is affine; replacing the power by its chord leaves a comparison of two sums linear in λ^t.

Theorem 7.3 (The chord criterion)**.** *For $\mu > 1$, if $W(R_2)D(R_1) \le W(R_1)\big(\iota\, D(R_1) + \varsigma\, D_\lambda(R_1)\big)$, where ι and ς are the intercept and slope of that chord and D_λ is the path price with each term discounted once more by λ^t, then initial consumption falls with the rate.*
Formal statement: *CertaintyEquivalent.consumption_antitone_of_chord*.

, CertaintyEquivalent.chord_le_pathPrice

The replacement costs almost nothing, because it is first-order exact: as $R_2 \to R_1$ it reproduces $T_y \ge (1 - 1/\mu)T_c$ exactly. At Aiyagari's numbers the chord threshold is 4.3263 against the exact 4.3267, and the cost grows only with the size of the rate step, to 0.04 at $\Delta r = 10^{-3}$ and 0.40 at 10^{-2} (`numerics/chord.m`). Together with the $\mu \le 1$ theorem this settles the certainty-equivalent household at every risk aversion.

The slack in the correction

The risk half of the criterion resists, for a reason set out in Appendix A: what has to be bounded is a difference of rate *responses*, and no bound on the level of consumption helps with that. But "resists" understates how good the position is, and the quantity worth measuring is not the correction but the room around it. Write the risky household's saving response at the state as the certainty-equivalent response less a correction, $\partial g/\partial r = \partial g_{\mathrm{CE}}/\partial r - \mathrm{corr}$. Theorem 1 at the state follows from any bound $B \ge \mathrm{corr}$ with $B \le \partial g_{\mathrm{CE}}/\partial r$ (`CertaintyEquivalent.saving_nonneg_of_correction_le`), so what matters is the ratio $(\partial g_{\mathrm{CE}}/\partial r)/\mathrm{corr}$: a bound loose by less than that factor still delivers the theorem. Measured at $r = 4\%$, zero wealth, top earnings state:

γ	1	2	2.5	3	3.2	3.39	5
$\partial g_{\mathrm{CE}}/\partial r$	20.98	7.28	4.55	2.73	2.17	1.69	−0.89
$\partial g/\partial r$ (risky)	17.08	4.88	2.49	0.92	0.44	0.03	−2.05
correction	3.89	2.40	2.06	1.82	1.73	1.66	1.15
slack factor	**5.4**	**3.0**	**2.2**	**1.5**	1.25	1.02	—

Two things stand out. The correction is *stable*: it stays between 1.15 and 3.89 across $\gamma \in [1, 5]$, and it *falls* as risk aversion rises, while the certainty-equivalent response collapses from 20.98 to below zero. The threshold is therefore set by the certainty-equivalent side and not by risk at all — which is the same conclusion the closed form gave, now with the correction measured rather than inferred. And the slack is generous where it matters: a bound on the correction may be wrong by a factor of 5.4 at $\gamma = 1$, 3.0 at $\gamma = 2$ and 1.5 at $\gamma = 3$ and still certify Theorem 1 there.

That is a far weaker demand than the one left open in Section 10, where the remaining estimate needs an absolute error below 0.03 on a quantity of size about one. Here a crude precautionary bound would do. Note only that the slack is *not* uniform in γ: the correction peaks at 3.89 at $\gamma = 1$ while $\partial g_{\mathrm{CE}}/\partial r$ is only 2.73 at $\gamma = 3$, so a single constant cannot cover the range — the bound must be allowed to depend on γ, which costs nothing, since Theorem 1 is wanted at a calibration and not as a family. Numerics: `riskgap.m`.

7.7 The precautionary split

The slack above is measured on derivatives at $r = 4\%$. Theorem 1 is a statement about an interval, so it is worth redoing in secant form, where the object to be bounded becomes concrete. Write the risky household's saving at the state as the certainty-equivalent saving plus precautionary saving, $g = g_{\mathrm{CE}} + P$ with $P = c_0^{\mathrm{CE}} - c_0 \geq 0$. Then Theorem 1 across $[r_1, r_2]$ is *exactly*

$$P(R_1) - P(R_2) \;\leq\; g_{\mathrm{CE}}(R_2) - g_{\mathrm{CE}}(R_1) \;=:\; \text{margin}, \tag{P}$$

a bound on how far precautionary saving falls across the interval (`CertaintyEquivalent.saving_le_of_precautionary_fall`). No derivative appears. Two weakenings of (P) are worth separating, and they close at different risk aversions. Measured over $r \in [4\%, 6\%]$ at zero wealth in the top earnings state:

γ	1	2	2.5	3	3.2	3.39
$P(R_1)$	0.109	0.154	0.176	0.198	0.206	0.214
$P(R_1) - P(R_2)$	0.044	0.032	0.028	0.025	0.024	0.023
margin	0.329	0.107	0.062	0.031	0.022	0.014
level route: $P(R_1) \leq$ margin	✓	×	×	×	×	×
θ admissible	—	0.305	0.650	0.842	0.895	0.936
θ true	0.598	0.793	0.839	0.872	0.883	0.892

At $\gamma = 1$ the crudest weakening works: discard $P(R_2) \geq 0$ and a bound on the *level* of precautionary saving suffices, with the margin 0.329 exceeding $P(R_1) = 0.109$ threefold (`saving_le_of_precautionary_level`). Above that the level route dies and one needs the multiplicative form: keeping a fraction θ of precautionary saving at the higher rate, (P) becomes $(1-\theta)P(R_1) \leq$ margin (`saving_le_of_precautionary_ratio`). At $\gamma = 2$ the admissible θ is 0.305 against a true ratio of 0.793: precautionary saving need only be shown not to lose seventy per cent of itself when the rate rises by two points, which is a crude statement. At $\gamma = 3$ the admissible θ is 0.842 against 0.872, a margin of three per cent. Above $\gamma \approx 3.1$ there is no margin at all, because Theorem 1 itself fails on that interval — the 3.39 of the derivative comparison is the threshold at $r = 4\%$, not across $[4\%, 6\%]$.

7.8 The level route closes at $\gamma = 1$

At $\gamma = 1$ the level route needs no new estimate at all. Precautionary saving is $P_1 = c_0^{\mathrm{CE}} - c_0$, so any *floor* $F \leq c_0$ bounds it above by $c_0^{\mathrm{CE}} - F$, and (P) reduces to one numerical inequality, $c_0^{\mathrm{CE}} - F \leq$ margin (`CertaintyEquivalent.precautionary_level_of_floor`). No bound on how precautionary saving moves with the rate is involved, and nothing at the higher rate.

The floor is already proved. `Incidence.riskHumanWealth_le_stageConsumption` gives $\kappa_k(m + H_k^{\mathrm{risk}}(z)) \leq c_k(z, a)$, with H^{risk} accumulated by the power mean of exponent $-\gamma$ — the harmonic mean at $\gamma = 1$ — and capped state by state at what the borrowing constraint permits. At zero wealth in the top earnings state across $r \in [4\%, 6\%]$:

required floor $c_0^{\mathrm{CE}} -$ margin	1.2176	$(= 1.5463 - 0.3287)$
$\kappa_{J-1}\, y_{\max}$	0.1303	no human wealth
$\kappa_{J-1}(y_{\max} + H^{\min})$	0.3966	`LowerSandwich`, 3.07× short
$\kappa_{J-1}(y_{\max} + H^{\mathrm{risk}})$	**1.3184**	`Incidence`, **clears it**
true c_0	1.4373	floor at 92%, 85% required

So $c_0^{\mathrm{CE}} - F = 0.2280 \leq 0.3287$, and Theorem 1 holds at that state, at logarithmic utility, across the bracket, from the constraint-incidence floor and the closed form — no new estimate. The route that had to be abandoned in Section 10 is the one that needs a tight bound on a rate *response*; this one needs only a tight bound on a *level*, and the programme already has one good enough.

7.9 The same result from two sandwiches

The certainty-equivalent apparatus turns out to be unnecessary for this. The quantity the level route needed a floor to beat, $c_0^{\mathrm{CE}} -$ margin, *is* certainty-equivalent consumption at the higher rate, and that in turn *is* the arithmetic upper sandwich $\kappa_k(m + H_k)$ — the deterministic propensity times resources plus the present value of the household's own expected earnings. So the $\gamma = 1$ result is just this:

Proposition 7.4 (Theorem 1 at a state, from two sandwiches)**.** *If the ceiling on consumption at the higher rate lies below the floor at the lower rate, consumption falls with the rate at that state.*

Formal statement: *CertaintyEquivalent.le_of_ceiling_le_floor.*

with C the arithmetic ceiling at R_2 and F the constraint-incidence floor at R_1. At zero wealth in the top earnings state across $r \in [4\%, 6\%]$:

γ	1	1.5	2	2.5	3
floor at R_1	1.3184	1.2672	1.2250	1.1886	1.1565
ceiling at R_2	1.2176	1.3495	1.4168	1.4576	1.4850
ceiling $\leq$ floor	✓	×	×	×	×
true $c_0(R_1) - c_0(R_2)$	0.2849	0.1452	0.0750	0.0332	0.0058
sandwich width lost	0.1841	—	0.2668	—	—

The failure at $\gamma = 2$ is structural, not a matter of sharpening constants. The width of the sandwich at a given rate *is* the precautionary wedge: the arithmetic human wealth gives the ceiling, the risk-adjusted one gives the floor, and they cannot both be tight, because the distance between them is exactly what precaution costs. So the route works while the consumption response to the rate exceeds that wedge, and no longer. Measured, the response is 0.285 against a wedge of 0.184 at $\gamma = 1$, and 0.075 against 0.267 at $\gamma = 2$ — short by a factor of 3.6, with both terms moving the wrong way as γ rises: the response collapses ($0.285 \to 0.075 \to 0.006$) while the ceiling and floor each lose accuracy ($1.057 \to 1.094 \to 1.131$ and $0.917 \to 0.894 \to 0.877$ as fractions of the truth).

Getting $\gamma = 2$ therefore needs what everything else in this programme needs: a bound on the *difference* between the two problems, rather than two bounds on the two problems separately. Two sandwiches cannot do it, however good, once the difference is smaller than their width.

Two things the $\gamma = 1$ result is not. It is one state — the state where the life cycle binds, which is why it is the one worth doing first, but Theorem 1 asks for all of them, and the reduction that would carry it across the state space is the one whose own condition fails. And it is $\gamma = 1$: the multiplicative route remains available in principle above that, needing $\theta \geq 0.305$ against a true 0.793 at $\gamma = 2$, but θ is a ratio of precautionary savings at two rates, which is again a difference between the two problems.

Two negative results are worth recording alongside. The iterated-harmonic formula $(y + H^{\text{harm}})/D$ is within 1.8% of the truth at the top state (1.4121 against 1.4373), and is *not* a floor: at the bottom state it gives 0.6588 against a true 0.2700, because it ignores the borrowing constraint, which is exactly what the cap in H^{risk} repairs. And Kantorovich's bound on the arithmetic-minus-harmonic gap accumulates to 84.4 against a true gap of 3.07; the exact variance identity $\text{AM} - \text{HM} \leq \text{Var}/x_{\min}$ does eight times better at 10.5, and neither is needed once the floor is used directly. Numerics: `precbound.m`, `logfloor.m`, `logPharm.m`, `g2.m`.

7.10 The channel is the borrowing constraint

The first-order scaling points at the constraint rather than at precaution, and the point can be tested directly: loosen the borrowing limit and watch the threshold. If the constraint is the channel, the risky threshold should walk back to the certainty-equivalent one as the limit approaches the natural one, $y_{\min}/r$. At Aiyagari's calibration it does (`numerics/borrow.m`), and it does so almost linearly:

Borrowing limit	0	0.5	1	2	3	4	5	6
Threshold μ	3.42	3.50	3.58	3.74	3.89	4.05	4.21	4.38
Share of the gap closed	0%	11%	21%	40%	57%	74%	90%	104%

The natural limit here is $y_{\min}/r = 6.75$, and by a limit of 6 the gap is gone: the risky household's threshold, 4.38, is the certainty-equivalent household's, 4.33. So at this calibration the borrowing constraint accounts for the whole of the correction, and precaution accounts for none of it. Precaution is certainly present — at the loose limit the risky household still saves 1.62 against the certainty-equivalent 1.46, a precautionary excess of eleven per cent — but it moves the level of saving without moving the sign of its response to the interest rate. Across the other calibrations we ran (`numerics/borrow3.m`) loosening the limit always closes the gap and sometimes overshoots it, so the constraint is the dominant channel rather than an exact decomposition: with ample borrowing, risk can make Theorem 1 easier than certainty equivalence rather than harder.

For the programme this is the useful part. The missing ingredient for $\mu = 3$ is not a precautionary bound but a bound on what the borrowing constraint does to the comparative static, and that is a more tractable object: without a binding constraint the household's Euler equation holds with equality at every state and Theorem 1 at zero wealth reduces to the duration criterion, which is proved outright for $\mu \leq 1$ and is an explicit finite computation in the primitives otherwise. What remains is the constrained states, which at this calibration are about eight per cent of the state space at every age.

7.11 The constraint channel is monotone, and needs no condition

The comparative static in the borrowing limit itself is a theorem, and unlike the one in the interest rate it needs nothing beyond the standing assumptions.

Theorem 7.5 (A tighter limit means less consumption and more saving)**.** *Two households share every primitive and differ only in the borrowing limit, $f_2 \leq f_1$. Then at every age and every state in the tighter household's region it consumes less and saves more.*
Formal statement: *IncomeFluctuation.finiteConsumption_le_of_floor_le, IncomeFluctuation.le_finitePolicy_of_floor_le.*

The induction is the one of Theorem 7.1 with the interest rate held fixed, and that is exactly what removes the difficulty. Suppose the claim holds with k periods to come and fails at $k+1$, so the tighter household consumes strictly more and therefore saves strictly less. Its Euler inequality under the cap, the looser household's Euler inequality above its floor —

available because the looser household saves strictly more than the tighter one and hence strictly more than either floor — the induction hypothesis, and monotonicity of consumption in wealth chain into $u'(c_1) \geq u'(c_2)$, contradicting $c_1 > c_2$. The gross return appears on both sides and cancels, so there is no substitution term to control and no analogue of (1) is needed. Tightening the limit raises the marginal value of saving unambiguously, and the terminal stage is again free: a household with a lower floor runs down further in its last period.

That is the one-sided half of what the numbers showed, and it says the constraint channel is better behaved than the rate channel rather than merely different. What it does not yet give is a *size*: Theorem 7.5 orders the two consumption functions without bounding the distance between them, and it is the distance that turns into the 0.9 units of risk aversion between the constrained and certainty-equivalent thresholds. The natural next object is therefore a modulus for Theorem 7.5, and it has a clean form.

Theorem 7.6 (The modulus: a relaxation of Δ is worth at most Δ of wealth)**.** *Let* $\Delta = f_1 - f_2 \geq 0$ *and* $r \geq 0$*. Then at every age and every state,*

$$c_k^{f_1}(x) \;\leq\; c_k^{f_2}(x) \;\leq\; c_k^{f_1}(x + \Delta).$$

Formal statement: *IncomeFluctuation.finiteConsumption_le_of_floor_le, IncomeFluctuation.finiteConsumption_le_shift.*

The household that may borrow down to f_2 therefore behaves, at wealth x, no more freely than the household confined to f_1 would at wealth $x + \Delta$: the entire value of the extra borrowing capacity is bounded by the capacity itself. The proof is the same backward induction, carried out at shifted arguments, and the shift pays for itself twice. The terminal stage holds with slack, since the looser household gains Δ by running down further while the shifted comparison gives it $(1 + r)\Delta$; and the interiority the step needs comes free, because the failure hypothesis forces the tighter household's saving to exceed the looser one's by more than Δ, hence to exceed $f_2 + \Delta = f_1$.

Two consequences. Since saving never falls with wealth, the sandwich gives the plain bound $0 \leq c_k^{f_2}(x) - c_k^{f_1}(x) \leq (1 + r)\Delta$, so the effect of the limit on consumption is at most the relaxation, up to one period's interest. And because the bound is stated as a shift in wealth rather than as an additive constant, it composes with the minimal propensity to consume: whatever upper bound on the propensity to consume out of wealth one has at age k, the modulus turns it directly into a bound on the consumption gap, and it is that product which becomes the gap between the constrained and certainty-equivalent thresholds.

7.12 The upper bound on the propensity to consume, and what it is worth

The missing piece is an upper bound on the propensity to consume out of wealth. The repository has the minimal propensity, which is a lower bound; the upper one follows by the same two-point Euler comparison, with no derivative taken. Write the bound as a secant slope in assets, so that s_k bounds $c_k(a_2) - c_k(a_1)$ by $s_k(a_2 - a_1)$.

Theorem 7.7 (The propensity to consume falls with the horizon)**.** *In the last period the slope is exactly R (`stageConsumption_sub_zero`). If at stage k the slope is at most s and consumption is at least $c_{\min}$, then at a state where the poorer household saves, the slope at stage $k+1$ is at most*

$$R \cdot \frac{\nu}{1+\nu}, \qquad \nu = \frac{c\,s}{c_{\min}},$$

with c the poorer household's consumption.
Formal statement: *`IncomeFluctuation.stageConsumption_sub_le_of_slope`.*

The step is the two Euler inequalities at the two wealth levels: the induction hypothesis bounds the rise in tomorrow's consumption by $s\delta$ with δ the difference in saving, which bounds the fall in marginal utility by the factor $(1 + s\delta/c_{\min})^{-\mu}$, and the budget identity $c_2 - c_1 = R(a_2 - a_1) - \delta$ turns that into the stated slope. Since $\nu \mapsto \nu/(1+\nu)$ is a contraction towards $1 - 1/\nu$, the slope falls from R towards $R(1 - c_{\min}/(c\,s))$ as the horizon lengthens: the longer the life, the less of an extra unit of wealth is spent at once, which is the qualitative fact one wants.

It is not sharp enough. At Aiyagari's calibration, read at zero wealth in the highest earnings state, consumption is about 1.32 and the worst consumption next period about 0.88, so $\nu \approx 1.5\,s$ and the recursion settles at a propensity of about 0.36, against a true propensity there of about 0.04. Fed through Theorem 7.6 it bounds the effect of the borrowing limit on consumption by about 2.5 where the truth is about 0.04, a factor of sixty. The loss is the ratio of consumption today to the worst consumption tomorrow, and that ratio is where the sandwich's slack of order human wealth enters — the same obstruction that closed the infinite-horizon routes in the companion write-up, met here for the fourth time and in the same form.

So the chain now has every link proved except its sharpness. Theorem 7.2 gives the criterion without risk, Theorem 7.5 signs the constraint's effect, Theorem 7.6 bounds it by a shift in wealth, and Theorem 7.7 converts a shift in wealth into consumption. What none of them supplies is a bound on consumption that is tight to a few per cent rather than to a factor of ten, and every route we have tried reduces to exactly that.

7.13 Carrying earnings across all remaining ages

The two halves of the sandwich are not equally to blame. The upper half, $c_k \le \kappa_k(m + H_k)$, carries the expected present value of future earnings and is about twenty per cent above the truth at Aiyagari's calibration. The lower half, $\kappa_k m \le c_k$, ignores future earnings entirely and is a factor of ten below it. The repair is the same device applied on the other side: carry earnings forward across all the remaining ages instead of one step at a time.

Theorem 7.8 (The lower half, with human wealth)**.** *Let H_k be the present value of k periods of the minimum earnings, $H_0 = 0$ and $RH_{k+1} = y_{\min} + H_k$. Then $\lambda_k(m + H_k) \le c_k(m)$ at every age and state, where $\lambda_0 = 1$ and $\lambda_{k+1} = \flat\lambda_k R/(\flat + \lambda_k R)$.*
Formal statement: *`IncomeFluctuation.minHumanWealth_le_stageConsumption`.*

The old bound is the case $H = 0$. The propensity λ_k is the finite-horizon propensity of Theorem 3.1 discounted once more by patience at each step, and the extra discount is not decoration: without it the bound is false at the borrowing constraint, where the household consumes exactly its cash on hand while $\kappa_k(m+H_k)$ exceeds it, because H_k discounts earnings at R and $1/\kappa_k$ discounts at $R/\text{þ}$. Discounting H_k at $R/\text{þ}$ instead repairs the corner and breaks the induction; the extra patience factor repairs both at once (`lowMPC_mul_minHumanWealth_le`), and at $\beta = 0.96$, $r = 4\%$ and $\mu = 3$ it costs about three per cent over sixty periods.

At the state where the life cycle binds the household enters the next period with cash on hand about 2.0 and consumes about 0.88. The old bound gives 0.086; this one gives 0.33. Through Theorem 7.7 that improves the bound on the propensity to consume from about 0.94 to about 0.75, against a true propensity of 0.04.

So the multi-age argument buys a factor of four and not a factor of ten, and the reason is visible in the statement: it carries the *minimum* earnings, and a household that drew the minimum forever really would consume this little. What the programme needs is a lower bound carrying the *expected* human wealth, and that is not available from any chain of Euler inequalities, because the Euler inequality at a constrained state is exactly the statement that expected future earnings cannot be borrowed against. That is the one quantitative fact about a single household on which the whole programme turns, and identifying it precisely is what makes it attackable.

7.14 The constraint-incidence bound

Both halves can be carried at once. Accumulate human wealth by the *risk-adjusted* expectation — the power mean with exponent $-\mu$, which is what the Euler equation itself produces — and then *cap* it state by state at the level the borrowing constraint permits:

$$H_0 = 0, \qquad H_{k+1}(z) = \min\Big\{ \tfrac{1}{R}\, M^z_{-\mu}\big(y_w + H_k(w)\big),\ \Big(\tfrac{1}{\kappa_{k+1}} - 1\Big) y_z \Big\}$$

(`riskHumanWealth`), with κ the propensity of Theorem 3.1.

Theorem 7.9 (The constraint-incidence bound)**.** *$\kappa_k\big(m + H_k(z)\big) \le c_k(a, z)$ at every age and state.*

Formal statement: *`IncomeFluctuation.riskHumanWealth_le_stageConsumption`.*

The two ingredients meet exactly where they should. The power mean is what the Euler inequality delivers, and the step needs only that it is superadditive under a shift, which follows from its concavity and homogeneity (`powerMean_add_const_le`). The cap is the constraint's incidence: at a corner the household consumes its cash on hand, which is at least y_z, and the cap is precisely the largest human wealth the bound may claim without exceeding that. No extra patience factor is needed, because the cap does the work the patience factor did in Theorem 7.8.

At Aiyagari's calibration, in the first period of a sixty-period life (`numerics/incidence.m`), the cap binds at the two lowest earnings states and nowhere else, which is where a household at zero assets is against the limit. The bound reaches

At zero wealth	lowest earnings	middle	highest
Bound as a fraction of the truth	1.000	0.890	0.877
Minimum-earnings bound, for comparison	1.010	0.409	0.295

and over the whole grid it never exceeds the truth, touching it exactly at the constrained corner. So the consumption bound is now within about twelve per cent where the life cycle binds, against a factor of three before. That is the sharpest bound in the development, and it was reached by bounding where the constraint binds rather than by another Euler chain.

It does not improve the propensity bound of Theorem 7.7, because that bound is driven by the worst earnings state next period, which is exactly where the cap binds. Closing the threshold argument would need the same sharpening on the worst state, and that turns out to be impossible within this class.

7.15 The worst state is pinned

Two facts settle it. The first is that the cap is not conservatism but necessity.

Theorem 7.10 (The cap is necessary)**.** *Wherever a household at zero assets is against the borrowing limit, every affine lower bound $\kappa_k(m+H)$ valid at that state obeys $H \leq (1/\kappa_k - 1)y_z$.*
Formal statement: *`IncomeFluctuation.cap_necessary_of_corner`.*

The proof is one line — at the corner consumption is exactly y_z — and the content is that Theorem 7.9 claims the largest human wealth any bound of its shape may claim. It cannot be sharpened within the affine class at all.

The second is that dropping the cap does not merely weaken the proof, it makes the bound false. Measuring effective human wealth as $c/\kappa - m$ at next-period assets of 1.67 (`numerics/worst.m`):

Effective human wealth at	truth	capped mean	uncapped mean	expectation
lowest earnings	10.96	5.99	13.11	16.57
highest earnings	27.92	23.90	26.38	32.24

At the lowest earnings state the uncapped power mean, 13.11, *exceeds* the truth, 10.96, so a bound built on it is invalid: the household will be against the limit later, and the uncapped recursion cannot see that. The truth sits between the two, and the distance is the price of capping at zero assets when the household in fact holds some. A bound that closed it would have to let human wealth depend on the assets held, which requires a lower bound on the household's own asset path; the one available, from the upper sandwich, is negative at the state that matters.

So the worst state is pinned from both sides: the cap cannot be raised, because the corner forbids it, and it cannot be dropped, because the truth is below what dropping it would claim. That closes this family of arguments, and with it, for now, the route from a consumption bound to the threshold. What the programme has instead is an exact statement of what a sharper argument must supply: a lower bound on consumption that is asset-dependent rather than affine, and the asset path to support it.

7.16 The bound with assets in it

The obstruction just described is no longer binding, because the path bound it asked for is available. Proposition 6.9 read at the household's own earnings state rather than at the worst one is positive exactly where the cap bites hardest: a household in a high earnings state saves, so a period later it is not at the corner, and the cap that then applies to it is the one for a household with assets.

So assets go through the recursion twice over. Writing $\Phi_k(z,a)$ for that saving floor, clamped at zero,

$$G_0(z,a)=0, \qquad G_{k+1}(z,a)=\min\Big(\tfrac{1}{R}M^z_{-\mu}(y_w+G_k(w,\Phi_{k+1}(z,a))),\ (\tfrac{1}{\kappa_{k+1}}-1)(y_z+Ra)\Big),$$

and the bound is $\kappa_k(m+G_k(z,a))\le c_k(a,z)$. The cap now grows with assets, and tomorrow's human wealth is read at the assets the household will hold rather than at zero.

Theorem 7.11 (The asset-dependent incidence bound)**.** *$\kappa_k(m+G_k(z,a))\le c_k(a,z)$, and $G_k(z,a)\ge H_k(z)$ for $a\ge 0$, so the bound is never worse than Theorem 7.9.*
Formal statement: *IncomeFluctuation.assetHumanWealth_le_stageConsumption.*

, IncomeFluctuation.riskHumanWealth_le_assetHumanWealth

Measured against the truth over the whole grid at three ages (`numerics/assetdep.m`, `numerics/worstgrid.m`):

Worst ratio of bound to truth	$r=-7\%$	$r=4\%$	$r=10\%$
Theorem 7.9, cap at zero assets	0.237	0.606	0.760
Theorem 7.11, cap at the assets held	0.350	0.746	0.857

The new bound never exceeds the truth anywhere on the grid. The gain is largest exactly where the old one was weakest, at positive assets in low earnings states: at $r=4\%$ with five units of assets and the lowest earnings, the ratio rises from 0.621 to 0.835. Where the old bound was already good, at zero assets, little changes, which is as it should be, since that is the state the cap was computed for.

What the refinement does not improve is the ceiling on aggregate capital, because the state that binds there is the highest earnings one, where the cap was never active: the implied ceiling at $r=-7\%$ is 22.15 against 22.17. That is still far below the 39.66 that feasibility alone gives, which is worth carrying into the existence argument of Section 6.

8 Theorem 1 at logarithmic utility

The slack condition of Section 6 is an Euler condition, and it is worth asking whether it settles the one case the stationary model does settle, $\mu=1$. It does not, and the reason is instructive. Checking it on the whole grid at $\gamma=1$ (`numerics/slacklog.m`), it holds at $r=0$ and $r=2\%$ and fails at $r=4\%$ and above, by about twelve per cent in elasticity terms, at the youngest age with zero assets in the highest earnings state. Since a life-cycle equilibrium

sits above the rate of time preference, 4.17% here, the Euler reduction says nothing where it is wanted, even at logarithmic utility.

8.1 A condition on the value gap

A different sufficient condition does hold. Write $V_k(\cdot; R)$ for the stage-k value.

Definition 8.1 (Value gap)**.** The value gap has slope at least θ at stage k if, for $x \le y$ in $[0, \bar{a}]$,

$$\theta\,(y - x) \;\le\; \big[V_k(y; R_2) - V_k(y; R_1)\big] - \big[V_k(x; R_2) - V_k(x; R_1)\big].$$

Formal statement: `IncomeFluctuation.ValueGap`.

By the envelope theorem the slope of the gap is $R_2 u'(c_2) - R_1 u'(c_1)$, so the condition says the marginal value of assets rises with the rate. At $\gamma = 1$ that reads $c(R_2)/R_2 \le c(R_1)/R_1$: consumption rises by proportionally less than the rate does.

Theorem 8.2 (Theorem 1 from the value gap)**.** *If the value gap has a positive slope at stage k, then saving at stage k rises with the rate, at every state. Aggregate capital rises with it.*
Formal statement: *`IncomeFluctuation.stagePolicy_mono_of_valueGap`.*

, `IncomeFluctuation.olgCapital_le_of_valueGap`

Unlike the Euler route this needs no induction over ages. The condition on the continuation gives the conclusion for the household that faces it, directly, and the argument is Topkis with two continuations: the two households differ only in their cash on hand, which is ordered by the rate, and in their continuations, whose difference is increasing, so the higher-rate household cannot save less. Only concavity of period utility is used, as in Section 5.

8.2 Why it holds at $\mu = 1$, and only near there

At $\gamma = 1$ the patience factor is $\mathfrak{p} = \beta R$, so $q = \mathfrak{p}/R = \beta$ and the propensity $\kappa_k = 1/\sum_{i \le k} \beta^i$ of Proposition 6.7 does not depend on the interest rate at all. Human wealth does, and falls as the rate rises. Where the borrowing constraint is slack, $c = \kappa_k(\bar{y} + Ra + H)$ and

$$R_2 c(R_1) - R_1 c(R_2) \;=\; \kappa_k\big[(R_2 - R_1)\bar{y} + R_2 H_1 - R_1 H_2\big] \;>\; 0,$$

which is the condition, with room. At a corner, $c = \bar{y} + Ra$ and the same inequality reduces to $R_1 \bar{y} \le R_2 \bar{y}$.

Numerically (`numerics/condB.m`) the condition holds at $\gamma = 1$ at every age, state and rate tested, and fails from $\gamma = 2$, where between 19% and 51% of states violate it. It is therefore sufficient and not necessary: Theorem 1 itself still holds at $\gamma = 2$, and up to about 3.4. So this route covers $\mu = 1$ and closes nothing above it.

What is proved, then, is the reduction, and it is a reduction to a condition that is true at logarithmic utility rather than one that is false there. Discharging the condition itself from primitives is not available from the sandwich of Theorem 3.1: the sandwich's width is $\kappa_k H_k$, and the rate response of exactly that quantity is what drives the gap, so the bounds cannot

see it. That is the same obstruction the consumption-bound family ran into above, and it is the piece a sharper argument would have to supply.

8.3 Light's induction is the stage induction

The sharper argument exists, and it is Light's own. His proof of Theorem 1 in the stationary model is an induction on Bellman iterates from zero: if a continuation's marginal value rises with the rate, so does the next iterate's, given concavity of the consumption function and relative risk aversion at most one. The iterates from zero are the stage values, so the induction is the stage induction, and the step — proved for the stationary model in `Models/IncomeFluctuationRateStep` — says exactly that the value gap of Definition 8.1 with $\theta = 0$ propagates from stage k to stage $k+1$. Two hybrid hypotheses of that step, positivity and cap slack of the higher-rate household's choice against the *lower*-rate continuation, are discharged inside the induction: positivity from unbounded utility, and slack from single crossing (the induction hypothesis) against the higher-rate household's own slack.

Theorem 8.3 (Theorem 1 at $\gamma \leq 1$, every calibration)**.** *Let utility be CRRA with $\gamma \leq 1$ and unbounded below at zero consumption, and let the saving cap be slack at the higher rate at every stage. Then the stage values at the two rates have increasing differences in wealth at every stage; saving rises with the rate at every state and age; aggregate capital rises with the rate; and a strictly falling demand meets it at most once. No condition on the discount factor, the earnings process or the horizon.*
Formal statement: *`IncomeFluctuation.stageValue_incDiffSlices`, `IncomeFluctuation.valueGap_zero_of_light`, `IncomeFluctuation.stagePolicy_mono_light_crra`, `IncomeFluctuation.olgCapital_le_light_crra`, `IncomeFluctuation.olgEquilibriumRate_unique_light_crra`.*

With the value gap at slope zero, Theorem 8.2 is not quite available — its proof breaks the tie with the margin θ — and the conclusion comes instead by the one-economy chain of the stationary file: the budget coincidence puts the lower-rate household's problem at (a, R_1) equal to the higher-rate household's at (ta, R_2), $t = R_1/R_2$, with the lower-rate continuation; single crossing in the continuation (which needs no margin, the lower maximiser being unique) moves to the higher-rate continuation; and monotonicity in wealth moves from ta to a.

Numerically the statement was already in hand: `numerics/condB.m` found the value gap non-negative at $\gamma = 1$ with a relative margin of 7×10^{-6} — a theorem with equality at the corners, as this one has — and violated from $\gamma = 2$. What the theorem adds is that the $\mu = 1$ uniqueness of Section 10 no longer rests on the certificate there: it holds at every calibration with $\gamma \leq 1$, by proof term. Under the standing requirement that utility be unbounded below at zero consumption, CRRA with $\gamma \leq 1$ means $\gamma = 1$ (`crra_pinch`); the theorem itself is stated for any $\gamma \leq 1$.

Nothing here reaches $\gamma > 1$. The pivotal inequality of the step needs $c\,u'(c)$ increasing, which is $\gamma \leq 1$ exactly, and the value gap itself is violated from $\gamma = 2$ on.

9 Mean field games, with age as time

The Lasry–Lions argument (Lasry and Lions, 2007) proves uniqueness in a mean field game from two optimality inequalities and a monotonicity condition on the cost. An earlier survey ruled it out for the stationary Bewley model, and for a stated reason: the argument cancels a term $\int (u_1 - u_2)\, d(m_1(0) - m_2(0))$ by assuming the two equilibria start from a common initial distribution, and a stationary discounted equilibrium has no common start — two candidate rates carry two different stationary distributions.

Overlapping generations do have one. Every cohort is born with nothing, whatever the rate, so the date-zero marginal is the same point mass crossed with the invariant earnings law at every candidate equilibrium. The stated obstruction is simply absent, and the monotonicity condition becomes a question to check rather than one to dismiss.

9.1 The game

Take age as time. A player is a cohort member, a path is $x = (a_0, \dots, a_J)$ with $a_0 = 0$, and the population is a distribution η over paths with date-t marginals m_t. The cost at date t is the negative of flow utility, with the budget substituted in,

$$F_t(x, m) \;=\; -\beta^t\, u\Big(w(K(m))y_t + \big(1 + r(K(m))\big)a_t - a_{t+1}\Big), \qquad K(m) = \int a\, dm,$$

so the population enters through the single scalar K, and through it through both factor prices. This is the price-through-an-aggregate structure of Graber and Matter (2024), except that the aggregate enters the budget rather than being added to the cost.

9.2 What the argument actually uses

Adding the two equilibrium inequalities — each equilibrium does no worse against its own costs than the other distribution does — signs the sum of the date pairings and nothing finer:

$$\sum_t \underbrace{\int \big(F_t(\cdot, m_{1t}) - F_t(\cdot, m_{2t})\big)\, d(m_{1t} - m_{2t})}_{P_t} \;\le\; 0.$$

Theorem 9.1 (The engine)**.** *Two relaxed equilibria of the finite-horizon game with a common initial distribution have $\sum_t P_t \le 0$. No monotonicity, no integrability and no continuity is used.*
Formal statement: *MeanFieldGame.summedPairing_nonpos_of_pathEquilibrium.*

The usual hypothesis, that every F_t is Lasry–Lions monotone, is one way to conclude from this and not the only one: it forces each summand to be zero. The weaker hypothesis that the argument can carry asks only that the total be nonnegative.

Theorem 9.2 (Uniqueness of an aggregate from the summed pairing)**.** *If the cost separates a statistic A of the marginals — $\sum_t P_t > 0$ whenever A differs — then two relaxed equilibria from a common initial distribution agree on A. No single date is assumed monotone.*

Formal statement: *MeanFieldGame.aggregate_eq_of_pathEquilibrium*.

Date-by-date monotonicity is the special case in which every summand is already non-negative (`summedLasryLionsMonotone_of_forall`). The distinction is not cosmetic here, as the next two subsections show: in this economy the date-by-date hypothesis is false and the summed one is not.

9.3 The price channel is monotone; the curvature is not

Suppose for a moment that u were affine. Then $F_t(x, m_1) - F_t(x, m_2) = -\beta^t\big[(w_1 - w_2)y_t + (r_1 - r_2)a_t\big]$ and the pairing is

$$P_t \;=\; -\beta^t(w_1 - w_2)\int y_t \, d(m_{1t} - m_{2t}) \;-\; \beta^t(r_1 - r_2)\int a_t \, d(m_{1t} - m_{2t}).$$

The first term is exactly zero: labour is exogenous and the earnings marginal at each age is the same invariant law in both equilibria. The second is $-\beta^t(r_1 - r_2)(A_{1t} - A_{2t})$, and the factor-price frontier makes r strictly decreasing in K, so it is strictly positive at every date at which the economy with more capital holds more assets. The interest channel is exactly the Lasry–Lions congestion term: more aggregate capital raises the cost of holding capital. The price channel of a production economy does not run the wrong way. It is the only part of the problem that is unambiguously right.

What breaks is the weighting. With u strictly concave the pairing is that same price term weighted by marginal utility, and the weights do not cancel, because the two economies choose different savings and so sit at different marginal utilities. Over all pairs of measures the damage is total: taking concave and convex envelopes in the aggregate shows that unrestricted monotonicity of this cost forces u to be affine. The question worth asking is therefore the restricted one — whether monotonicity holds over the distributions the model can actually generate.

9.4 What the measurements say

Two facts settle the shape of what follows; the tables are in Appendix A.

Date by date the condition is false, at every risk aversion including logarithmic — the pairing is negative over the first three to six dates of life — and it survives refinement, with $P_0/(\Delta\beta)^2$ converging to -25.6 at $\gamma = 1$ and -44.4 at $\gamma = 3$. The failure is worst at birth, and that is the telling part: every cohort is born with nothing, so the affine pairing of the previous subsection is identically zero there, the aggregate coupling having nothing to price. The common start does not make the date-zero pairing vanish; it removes the one term that could have made it positive.

The summed condition is another matter. Over the family the equilibria belong to, it holds throughout the bracket $K \in [3.55, 7.40]$ in which Theorem 6.18 places the equilibrium, at $\gamma = 1, 2, 3, 5$, with a least margin that *grows* with risk aversion, from 0.075 to 0.338. So the weakening is not cosmetic: the hypothesis the argument can use is true here and the one usually imposed is not.

9.5 The reduction to submodularity

The summed condition is an integral inequality over pairs of measures. It is not: it is a submodularity condition on a function of two real arguments, and the reduction is exact.

Let $U_i(K)$ be the lifetime utility of the *plan* carried by m_i — its asset path — when that plan is priced at aggregate K. Expanding the definitions and collecting terms,

$$\sum_t P_t \;=\; -\Big[\,U_1(K_1) - U_2(K_1) \;-\; U_1(K_2) + U_2(K_2)\,\Big],$$

a mixed difference. (Checked numerically: the two sides agree to 10^{-14} in every case reported above.) So summed Lasry–Lions monotonicity is exactly the statement that $U(K, \text{plan})$ has *decreasing differences* in the aggregate and the plan: the plan that goes with the larger aggregate loses its relative advantage as the aggregate rises.

Theorem 9.3 (The reduction)**.** *When the population enters the cost only through a real statistic θ, the summed pairing is the mixed difference of the whole-life cost in (population, statistic). Strictly increasing differences of that cost make θ the same at every relaxed equilibrium with a common start.*
Formal statement: *`MeanFieldGame.famSummedPairing_scalar`.*

`, MeanFieldGame.statistic_eq_of_increasingDifferences`

This needed one repair. A stationary overlapping-generations aggregate is an average across *all* ages, so the date-t cost depends on the whole family of marginals and not on the date-t marginal alone. The finite-horizon machinery has been redone in that generality (`famPathCost`, `IsFamPathEquilibrium`, `famSummedPairing`, `famSummedPairing_nonpos_of_pathEquilibrium`); the engine survives unchanged.

The two channels

Differentiate. For *any* constant-returns technology, Euler's theorem gives $\partial w/\partial K = -(K/L)\,\partial r/\partial K$, so with L normalised to one,

$$\frac{\partial U}{\partial K}(K, \text{plan}) \;=\; r'(K)\,\Phi(K, \text{plan}), \qquad \Phi(K, \text{plan}) \;=\; \sum_t \beta^t\,\mathbb{E}[u'(c_t)\,(a_t - K y_t)],$$

where c_t is the plan's consumption *at the pricing aggregate* K. The quantity $a_t - K y_t$ is the household's **net capital position**: its assets less the capital that goes with its own labour at the economy's capital–labour ratio. Since $r' < 0$, decreasing differences say

$$\Phi(K, \text{plan}_1) \;\geq\; \Phi(K, \text{plan}_2) \qquad \text{at every common } K, \tag{W}$$

for plan_1 the plan carrying the larger aggregate: *the plan with more capital has the larger marginal-utility-weighted net capital position.*

9.6 The weighted net capital position

Two things can be proved. Neither closes (W), and the second says why.

The duration identity

Along the diagonal, where a plan is priced by the aggregate it was chosen for, Φ has a closed form. Let $m_t = \beta^t u'(c_t)$. Interiority makes m a state-price density, $\mathbb{E}_t[m_{t+1}] = m_t/R$, and writing $P[x] = \sum_t \mathbb{E}[m_t x_t]$ and $P_1[x] = \sum_t t\,\mathbb{E}[m_t x_t]$, two identities follow from the budget recursion $a_{t+1} = Ra_t + wy_t - c_t$ with $a_0 = a_{J+1} = 0$: the lifetime budget in state-price units, $P[wy - c] = 0$, and $P[a] = -P_1[wy - c]/R$.

Proposition 9.4 (Φ as a duration wedge)**.** *For the plan optimal at K, with $Y = P[y]$ and $T_c = P_1[c]/P[c]$, $T_y = P_1[y]/P[y]$ the marginal-utility-weighted durations of consumption and of earnings,*

$$\Phi(K, p(K)) \;=\; Y\Big[\tfrac{w}{R}\,(T_c - T_y) \;-\; K\Big].$$

Assets are the wedge between the duration of spending and the duration of earning; the household's marginal valuation of a higher interest rate is negative exactly when that wedge, scaled by the wage, falls short of the economy's capital–labour ratio. At the calibration it does, at every risk aversion: $T_c - T_y$ runs from 1.7 to 3.8 years while $K \approx 5$, so $\Phi < 0$ — along the factor-price frontier a higher interest rate *hurts* the household, because $dw/dr = -K$ and the marginal-utility weighting favours the young, who have no assets to gain on. Checked numerically: $P[wy-c]/P[c] = 3\times10^{-5}$, $P[a]+P_1[wy-c]/R = 1.7\times10^{-2}$ against $P[a] = 57.9$, and the two sides of Proposition 9.4 agree to one part in 10^4; the residual is the borrowing constraint, which binds on 3–11% of the mass and breaks the Euler equality there. At $\gamma = 1$ the identity $m_t c_t = \beta^t$ makes $T_c = \sum_t t\beta^t / \sum_t \beta^t$ exactly, independent of prices and of the earnings process.

This is the diagonal, and (W) is about two plans priced at a *common* aggregate that is neither's own. The identity does not reach it.

The obstruction

The reason (W) resists is not that the argument is delicate. It is that the inequality is the whole of uniqueness, and the following says so.

Proposition 9.5 (The pairing is minus two revealed-preference losses)**.** *For any cost, the summed pairing satisfies the identity $\sum_t P_t = -[(\mathcal{C}_1[m_2] - \mathcal{C}_1[m_1]) + (\mathcal{C}_2[m_1] - \mathcal{C}_2[m_2])]$, where $\mathcal{C}_i[m]$ is the whole-life cost borne by population m at the prices set by population i. At two relaxed equilibria with a common start both brackets are nonnegative.*

Formal statement: *MeanFieldGame.famSummedPairing_eq_neg_losses.*

, *MeanFieldGame.famCost_le_of_pathEquilibrium*, *MeanFieldGame.famSummedPairing_nonpos_of_losses*

At two *distinct* equilibria both brackets are strictly positive, so $\sum_t P_t < 0$ strictly. A hypothesis forcing $\sum_t P_t > 0$ is therefore not a statement about nearby populations that could be established by a local expansion and then extended: it is, on the nose, the assertion that no second equilibrium exists. That kills the route we would otherwise have taken — expand the optimality loss to second order about the diagonal, where it is a positive quadratic, and

read off a radius within which the sign is guaranteed. There is no such radius, because the sign that a local expansion delivers at a hypothetical pair of equilibria is the wrong one. (W) is strictly stronger than uniqueness, as Light's Theorem 1 also is, and any proof of it has to be global.

Optimality gives the differences falling for free over the interval between the two *planning* aggregates — add the two optimality inequalities — while what is needed is the same fall between the two *carried* aggregates; the intervals coincide only at equilibrium, and there Proposition 9.5 turns the free inequality against us.

9.7 Comonotonicity, and the collapse

Since no local argument can work, the proof has to be global. Pushing it there settles the route, though not in the direction we wanted.

Re-pricing a plan, exactly

One piece of structure is worth recording because it is exact and it is the natural starting point for any global argument. Parametrise by the rate rather than the aggregate. The factor-price frontier is $dw/dr = -K(r)$, so for a plan with asset path a chosen at rate $\hat{r}$ and re-priced at rate r,

$$w(r) - w(\hat{r}) = \int_{\hat{r}}^{r} w'(s)\,ds = -(r-\hat{r})\,\bar{K}, \qquad \bar{K} := \frac{1}{r-\hat{r}}\int_{\hat{r}}^{r} K(s)\,ds,$$

and therefore, at every date and in every state,

$$c_t(r) \;=\; \hat{c}_t \;+\; (r-\hat{r})(a_t - \bar{K}y_t). \tag{R}$$

Re-pricing shifts consumption by the rate gap times the net capital position *at the average capital intensity over the interval*. It is an identity, not an approximation. One consequence is immediate: differentiating Φ in the pricing gap at a fixed plan gives $\sum_t \beta^t \mathbb{E}[u''(c_t)(a_t - \bar{K}y_t)(a_t - Ky_t)]$, which is negative wherever the two net positions agree in sign — everywhere outside a band of asset-to-earnings ratios of width $|K - \bar{K}| = O(r-\hat{r})$. So Φ falls in the pricing gap at a fixed plan. What (W) needs is the comparison across *different* plans, and (R) does not supply it.

The weakest form of the hypothesis

Before grinding further it is worth asking what the least one could get away with is. The uniqueness chain runs: two stationary equilibria (K_1, r_1), (K_2, r_2) with $K_1 > K_2$; the firm's demand is strictly decreasing, so $r_1 < r_2$; the hypothesis gives $\Delta^2 \le 0$; revealed preference gives $\Delta^2 > 0$ by Proposition 9.5; contradiction. Nothing in that chain asks capital supply to be monotone in the rate. All it asks is

$$A(\hat{r}_1) > A(\hat{r}_2) \implies \int \left[\Phi(\cdot, p(\hat{r}_1)) - \Phi(\cdot, p(\hat{r}_2))\right] \;\ge\; 0, \tag{C}$$

that the carried aggregate A and the weighted net position Φ be *comonotone* along the family of optimal plans. Neither is required to be monotone in its own right. That is genuinely weaker in kind than the single-crossing route, which needs A itself increasing — the step this programme cannot take above $\mu \approx 3.4$.

The collapse

It is weaker in kind, and it is not weaker here.

Proposition 9.6 (Comonotonicity collapses to monotone supply)**.** *If A and Φ are comonotone along the family and Φ is strictly monotone along it, then A is monotone along it.*
Formal statement: *`MeanFieldGame.monotone_of_comonotone_of_strictMono`.*

And Φ *is* strictly monotone along the family: that is measured in Appendix A, at every one of thirteen steps across $\hat{K}/K^* \in [0.5, 2]$, at every pricing point and every risk aversion. So (C) delivers monotone capital supply — and monotone capital supply already gives uniqueness by single crossing, with no mean field game anywhere in the argument. The Lasry–Lions route, taken to its end, is not logically weaker than the route it was meant to bypass.

This is a definite closure rather than an open problem, and it is worth being exact about what it does and does not say. It does not say (W) is false: every measurement says it is true. It says that a proof of (W) would be a proof of monotone capital supply, so there is no shortcut hiding in the mean field formulation. The one escape is an economy in which Φ is *not* monotone along the family, where (C) would be strictly weaker than monotone supply; we have not found one, and at this calibration none exists.

9.8 What this leaves

Three things, in order of how much they are worth.

The negative result is definite and it is not the expected one. Date-by-date Lasry–Lions monotonicity fails in this economy, for every risk aversion including logarithmic, and the reason is not that the price channel runs backwards — it runs the right way, exactly — but that strict concavity reweights it, most severely at the ages where there are no assets to price. This is not a $\mu = 1$ frontier of the kind the rest of the programme keeps running into; it fails at $\mu = 1$ too.

The structural gain is real. Theorem 9.1 isolates what the argument does without any hypothesis, and Theorem 9.2 turns the summed condition directly into uniqueness of the aggregate, which for this economy is uniqueness outright: one K means one r, one household problem, one answer. The route is no longer dismissed for the wrong reason.

What is missing is a proof of (W) from primitives, and Proposition 9.6 says why it will not come cheaply: a proof of it would be a proof of monotone capital supply. The condition is strictly weaker than the *pointwise* policy ordering — it survives at $\mu = 5$ where that fails, and its measured margin improves as risk aversion rises — but it is not weaker than the aggregate monotonicity that the single-crossing route needs. As a route to uniqueness this one is closed. What survives it is the reduction itself: the summed pairing is a mixed difference

(Theorem 9.3), Φ is a duration wedge (Proposition 9.4), and re-pricing is exact (R) — all of which are statements about the life-cycle household rather than about mean field games, and none of which depended on the route working.

10 The increment of capital supply

Section 9 ends by closing a route and leaving the original target: uniqueness through single crossing, which the programme has pursued through Light's Theorem 1 and cannot get above $\mu \approx 3.4$. This section changes what has to be proved. Theorem 1 need not hold. It need only fail on little mass.

10.1 What single crossing actually asks

Work in wage units, where homotheticity makes capital supply per unit of wage a function of the rate alone and clearing reads $S(r) = D(r)$ with $D(r) = \alpha/((1-\alpha)(r+\delta))$. Excess supply is increasing — which is all single crossing needs — as soon as

$$S(r_2) - S(r_1) \;>\; D(r_2) - D(r_1), \qquad r_1 < r_2, \tag{I}$$

that is, as soon as supply does not fall *faster* than demand. Demand is steep at the calibration, and that is the whole opportunity: over the proved bracket $r \in [2\%, 8\%]$ supply's worst secant slope is $+33$ at $\gamma = 5$ against -54 for demand, and it never bends backwards at any γ up to 12. A proof of (I) may be lossy by more than the true increment and still close.

The two-sided bracket on the *level* of capital that the existence theorem supplies cannot be used for this. As $r_2 - r_1 \to 0$ the bracket-only condition $F(r_2) - C(r_1) > D(r_2) - D(r_1)$ degenerates into $F \geq C$, a bracket of zero width; measured, its margin runs from -352 to -1737. The object to bound is the increment, not the level twice.

The two-sided bracket on the *level* of capital that the existence theorem supplies cannot be used for this, for a reason that recurs below and is set out in Appendix A: a bracket as wide as the quantity being bounded contributes a spurious spread larger than the real one.

10.2 Coupling the two economies

Run both economies on a common earnings path from the common start $a_0 = 0$, and let $x_j = a_j(r_2) - a_j(r_1)$. Split the step at the *lower*-rate state:

$$x_{j+1} = \underbrace{\left[g_j^2(a_j^2) - g_j^2(a_j^1)\right]}_{\text{same policy, different assets}} + \underbrace{\left[g_j^2(a_j^1) - g_j^1(a_j^1)\right]}_{\text{same state, different rate}}.$$

The second bracket is the failure of Theorem 1 at that state, which (`stagePolicy_le_iff_consumption_le`) is exactly $c^2 - c^1 \leq (r_2 - r_1)a$ read with a slack; call its shortfall e_j. The first is at least $-L_j\,(x_j)^-$ when saving is nondecreasing in assets and Lipschitz with constant L_j. So

$$x_{j+1} \;\geq\; -L_j\,(x_j)^- \;-\; e_j. \tag{$*$}$$

Nothing in $(*)$ asks $x_j \geq 0$: the pointwise ordering of policies is not assumed, and its failure is priced by e_j rather than excluded.

Theorem 10.1 (The gap envelope)**.** *If $x_0 \geq 0$, $L_j \geq 0$, $e_j \geq 0$ and $(*)$ holds at every age, then $x_j \geq -f_j$ where $f_0 = 0$ and $f_{j+1} = L_j f_j + e_j$; hence $\sum_j x_j \geq -\sum_j f_j$. With demand falling by more than the average of f,* (I) *follows.*
Formal statement: *OLG.gapAccum.*

, OLG.negPart_le_gapAccum, OLG.neg_gapAccum_le, OLG.sum_neg_gapAccum_le, OLG.lt_of_increment_bound

10.3 The mass-weighted form

The negative part is the point of the formulation. From $(*)$,

$$(x_{j+1})^- \;\leq\; L_j\,(x_j)^- + e_j,$$

whose right-hand side is *linear* in the two quantities. Expectations therefore pass straight through it, and averaging over the cohort replaces e_j by the **mass-weighted** failure of Theorem 1, $\bar{e}_j = \mathbb{E}_1[(g_j^1 - g_j^2)^+]$, in place of its worst case over states. Since the violation is confined to a sliver of the distribution, that is the difference between a bound that fails and one that passes by two orders of magnitude.

Measured at the calibration, with $L_j = (1-\kappa_k)(1+r_2)$ from the sandwich and the accumulation $\Gamma = \frac{1}{J}\sum_j f_j/\varepsilon$ around 21 to 24:

	$r \in [4\%, 6\%]$, budget 0.670		$r \in [2\%, 8\%]$, budget 2.109	
γ	5	8	5	8
worst violation ε	0.0460	0.0701	0.1313	0.2053
mass carrying it	0.0091	0.0666	0.0084	0.0636
bound, uniform ε	0.976 ×	1.464 ×	3.125 ×	4.707 ×
bound, age-dependent ε_j	0.760 ×	1.218 ×	2.498 ×	3.953 ×
bound, mass-weighted $\bar{e}_j$	**0.0046** ✓	**0.0310** ✓	**0.0138** ✓	**0.0958** ✓
true $S(r_2) - S(r_1)$	+0.692	+0.457	+2.060	+1.368

At $\gamma \leq 3$ every column is zero: in wage units Theorem 1 holds exactly, so $\varepsilon = 0$ and the bound is trivial. The interest is at $\gamma = 5$ and 8, where Theorem 1 fails and the mass-weighted bound passes with factors of 145 and 22 to spare — and passes on the wide bracket too, where the rates differ by six percentage points. Charging the worst violation at every age fails; charging it where it occurs does not. Age-dependence alone does not rescue the worst-case version, because the violation is spread over 53 of 60 ages while being tiny at nearly all of them.

10.4 The Lipschitz constant

The engine is proved, and so is the Lipschitz constant it runs on.

The propensity floor

Theorem 10.2 (The propensity to consume is at least κ_k)**.** *At every stage, every income state and every pair of asset levels in the region,*

$$\kappa_k\,(1+r)(a_2-a_1)\ \le\ c_k(z,a_2)-c_k(z,a_1),\qquad a_1\le a_2,$$

so saving is Lipschitz in assets with constant $(1-\kappa_k)(1+r)$*, which is the* L_j *the increment argument uses.*
Formal statement: *`IncomeFluctuation.stageMPC_mul_le_stageConsumption_sub`.*

The terminal condition is an identity rather than an assumption: in the last period consumption is cash on hand, so the propensity is exactly $1=\kappa_0$. The step is the two-point Euler comparison. With b_i the saving at a_i and $t=\kappa_k R(b_2-b_1)\ge 0$, the induction hypothesis raises next period's consumption by at least t in *every* state, so the Euler equation above the floor at a_2 gives

$$c_2^{-\gamma}\ \le\ \beta R\sum_{z'}\pi\,\big(C'(z')+t\big)^{-\gamma}\ =\ \beta R\,M(C'+t)^{-\gamma}\ \le\ \beta R\,\big(M(C')+t\big)^{-\gamma},$$

with M the certainty equivalent, the power mean of exponent $-\gamma$. The Euler equation under the cap at a_1 gives $M(C')\ge \text{Þ}c_1$, and $\beta R=\text{Þ}^{\gamma}$ turns the chain into $c_2\ge c_1+t/\text{Þ}$, which is the claim at stage $k+1$ once denominators are cleared.

The last inequality in the display is **superadditivity of the certainty equivalent**, $M(x+t)\ge M(x)+t$, which is concavity and homogeneity of degree one at the midpoint (`powerMean_add_const_le`, already proved for the constraint-incidence bound of Section 7). It is worth pausing on: this is the direction a negative power mean obliges, and it is the reverse of what several arguments in this programme wanted and could not have. The recurring obstruction, read the other way round, is what makes the floor go through.

Measured, the floor holds at every stage and every reached state with equality only at the last stage, and the constant is nearly tight: the true accumulation is 20.7 against the sandwich's 21.2.

The Lipschitz bound it gives

Theorem 10.3 (Accumulated assets are Lipschitz in assets)**.** *For* $a_1\le a_2$ *in the region,*

$$A_k(a_2,z)-A_k(a_1,z)\ \le\ G_k\,(a_2-a_1),$$

with G_k *the slope* `cohortSlope` *of the affine ceiling:* $G_0=1$, $G_{k+1}=1+G_k(1-\kappa_{k+1})(1+r)$.
Formal statement: *`IncomeFluctuation.cohortAssets_sub_le_cohortSlope`.*

The induction is the one already written for the affine floor, with one substitution: the propensity floor turns a step of assets into a step of saving no larger than $(1-\kappa_{k+1})(1+r)$ times it, and the expectation over next period's earnings passes the constant through unchanged. The same G_k that bounds the *level* of accumulated assets in Section 6 therefore bounds their

differences, which is what the increment argument accumulates — and it is the L_j whose accumulation was measured at 21 above.

10.5 The mass-weighted failure

What is left is $\bar{e}_j$, an expectation against a distribution nobody controls. It looks like the hard part, and it would be if the failure of Theorem 1 were spread over assets. It is not.

Measured, the failure of Theorem 1 is not spread over assets but confined to the high-earnings states — at $\gamma = 5$ the top two of seven, carrying invariant mass 0.095. That is the coordinate in which no estimate is needed.

Why that coordinate is free

Earnings are exogenous and the cohort is born with its state drawn from the invariant law, so the earnings marginal is ν at *every* age, exactly. Nothing has to be estimated in that coordinate.

Theorem 10.4 (The earnings marginal of a cohort is ν)**.** *If ν is invariant for π, the ν-weighted expectation of a function of the earnings state is unchanged by any number of steps of the cohort recursion. Hence $\bar{e}_j \le \sum_z \nu_z V_j(z)$ with $V_j(z)$ the supremum over assets of the failure at earnings state z: exact in earnings, worst-case only in assets.*
Formal statement: *IncomeFluctuation.incomeStep.*

, IncomeFluctuation.stageStep_of_incomeFn, IncomeFluctuation.sum_mul_incomeStep, IncomeFluctuation.sum_mul_incomeStep_iterate, IncomeFluctuation.sum_mul_le_of_le

Being exact in the coordinate where the failure lives is what makes the bound work:

	$r \in [4\%, 6\%]$, budget 0.670		$r \in [2\%, 8\%]$, budget 2.109	
γ	5	8	5	8
worst-case in both coordinates	0.976 ×	1.464 ×	3.125 ×	4.707 ×
Markov in assets, best threshold	0.976 ×	1.464 ×	3.125 ×	4.707 ×
ν-exact, worst-case in assets	**0.032** ✓	**0.130** ✓	**0.103** ✓	**0.422** ✓
fully mass-weighted (the truth)	0.0046	0.0310	0.0138	0.0958

The ν-exact row passes with factors of 21, 5.1, 20 and 5.0, and it is within a factor of seven of the fully mass-weighted truth — the whole of the remaining looseness being the worst case over assets inside each earnings state.

10.6 The two-rate difference

One quantity remains: $V_j(z)$, the largest failure of Theorem 1 over assets at a single earnings state. There is no distribution in it, and the accumulation leaves an allowance of $0.670/21 \approx 0.032$ per age for $\sum_z \nu_z V_j(z)$. Two routes to it suggest themselves and both fail — a quantitative version of the Euler step, which amplifies, and a direct bound from the sandwich,

which is an order of magnitude too loose (Appendix A). What works is a different chaining of the same two Euler relations.

Chaining through the certainty equivalent

That amplification is not intrinsic to comparing the two rates; it is an artefact of how the expectation was bounded. The minimum-based step (Q) of Appendix A replaces $\sum \pi\, C^{-\gamma}$ by its smallest term, which charges every stage's error against next period's consumption in the *worst* earnings state. Chaining the same two Euler relations through the certainty equivalent instead leaves no minimum anywhere:

Proposition 10.5 (Certainty-equivalent chaining)**.** *From* $c_2^{-\gamma} \geq K_2 M_2^{-\gamma}$ *and* $c_1^{-\gamma} \leq K_1 M_1^{-\gamma}$,

$$\frac{c_2}{c_1} \;\leq\; \Big(\frac{K_1}{K_2}\Big)^{1/\gamma} \frac{M_2}{M_1},$$

with M_i *the certainty equivalent of next period's consumption.*
Formal statement: *`CertaintyEquivalent.mul_le_mul_of_euler_pair`.*

Propagating a constant shift through M then costs the power mean's gradient sum $A = \sum_{z'} \pi\,(C/M)^{-\gamma-1}$, which is 1 exactly when next period's consumption is riskless. So the per-stage multiplier becomes a measure of the dispersion of next period's consumption under the risk-tilted measure, rather than the ratio of today's consumption to its worst realisation:

γ	1	2	3	5	8
multiplier, minimum-based	5.75	5.47	5.20	4.72	4.20
multiplier A, certainty-equivalent	**1.041**	**1.063**	**1.086**	**1.132**	**1.189**
qA, with $q = (R_1/R_2)^{1/\gamma}$	1.022	1.053	1.079	1.127	1.186
$(qA)^{59}$	3.6	21	88	1180	2.2×10^4

A fivefold reduction per stage, and at $\gamma = 1$ the sixty-stage accumulation falls from 5.75^{59}, which is beyond any use, to 3.6. It is still not a contraction: $qA > 1$ throughout, so the recursion grows, and the per-stage criterion it yields, $b_1 \leq (\text{þ}/qA)(a + c_1/(\gamma R))$ — which is the `StageSlack` of Section 7 with the risk penalty A made explicit — still fails by fourteen per cent at zero wealth in the top earnings state at $\gamma = 1$.

The propensity floor damps the step

The factor of 1.02 closes once the propensity floor is put into the step, where it belongs. Under the failure hypothesis the household at the lower rate saves MORE, by exactly the failure D, and saving more raises next period's consumption by at least $\kappa_k R_1 D$ (Theorem 10.2). Carrying that through,

$$D\Big[1 + \frac{qA\kappa_k R_1}{\text{þ}_1}\Big] \;\leq\; \frac{qA}{\text{þ}_1}\varepsilon_k + (q-1)c_1 + \frac{qA}{\text{þ}_1}\Delta r\, b_2 - \Delta r\, a,$$

so the error carried from one stage to the next is multiplied not by $qA/þ_1$ but by

$$m_k \;=\; \frac{qA/þ_1}{1+qA\kappa_k R_1/þ_1} \;=\; \frac{qA}{þ_1+qA\kappa_k R_1},$$

at most one exactly when $qA \le þ_1 + qA\kappa_k R_1$ (`CertaintyEquivalent.damped_le_one`). The damping is the propensity floor feeding back against the failure.

γ	1	1.5	2	2.5	3	5
$\max_k m_k$	**0.978**	**0.992**	1.002	1.012	1.021	1.035
$\prod_k m_k$	0.027	0.062	0.112	0.196	0.337	0.785

At $\gamma \le 1.5$ the recursion contracts stage by stage. And the product over the sixty stages is below one at *every* risk aversion tested, because κ_k approaches one late in life, where the damping is strongest — so an error carried from the terminal condition, which is exactly zero, is damped over a life even at $\gamma = 5$. This is the first non-amplifying propagation of the two-rate difference in this programme.

The multiplier is a variance

The multiplier A has to be bounded from primitives, and what does that is the exact constraint M carries rather than any bound on the spread of C (Appendix A collects the attempts that use the spread). Writing $u = (C/M)^{-\gamma}$ for the risk tilt, the definition of the power mean says $\mathbb{E}_w[u] = 1$ *exactly*, and the gradient sum is $A = \mathbb{E}_w[u^{1+1/\gamma}]$. At $\gamma = 1$ the exponent is two and

Proposition 10.6 (The gradient sum is one plus the variance)**.** *With* $\mathbb{E}_w[u] = 1$*,* $\mathbb{E}_w[u^2] = 1 + \mathrm{Var}_w(u)$.
Formal statement: *`sum_mul_sq_eq_one_add_variance`.*

so $A = 1 + \mathrm{Var}_w(u)$, with no inequality anywhere. Above $\gamma = 1$ the exponent falls below two and Lyapunov gives $A \le (1 + \mathrm{Var}_w(u))^{(\gamma+1)/(2\gamma)}$.

γ	1	1.5	2	2.5	3
allowance on A	1.066	1.060	1.056	1.054	1.053
A true	1.041	1.050	1.058	1.067	1.076
$\mathrm{Var}_w(u)$	0.041	0.092	0.165	0.270	0.416
bound from the variance	**1.041**	1.076	1.122	1.182	1.261
slack on the variance	**1.61**×	0.78×	0.46×	0.29×	0.19×

At $\gamma = 1$ the bound is exact and leaves 1.61 times of room on the variance at the newborn stage — 2.35 times at stage twenty — so the whole chain closes there on one remaining primitive input: $\mathrm{Var}_w((C/M)^{-1}) \le 0.066$, against a measured 0.041. Above $\gamma = 1$ it does not: Lyapunov loses accuracy as the exponent falls below two, and the variance itself grows tenfold by $\gamma = 3$, so the bound exceeds the allowance from $\gamma = 1.5$ on.

That Lyapunov step interpolates between the zeroth and second moments and ignores that the first moment is exactly one. Interpolating between the first and second instead gives the sharp bound.

Proposition 10.7 (The sharp moment bound)**.** *For weights summing to one, $u \geq 0$ with $\mathbb{E}_w[u] = 1$ and $1 < p < 2$, $\mathbb{E}_w[u^p] \leq (\mathbb{E}_w[u^2])^{p-1}$, with equality for the two-point law on $\{0, \mathbb{E}_w[u^2]\}$. With $p = 1 + 1/\gamma$: $A \leq (1 + \mathrm{Var}_w(u))^{1/\gamma}$.*
Formal statement: *sum_mul_rpow_le_rpow_sum_mul_sq.*

It is Jensen for the concave $x \mapsto x^{p-1}$ under the tilted weights $w_i u_i$, which sum to one by hypothesis; or, in the language of the classical moment problem, the lower principal representation of the Chebyshev system $\{1, x, x^2\}$ for x^p with $p \in (1, 2)$. The exponent $1/\gamma$ is below Lyapunov's $(\gamma + 1)/(2\gamma)$ for every $\gamma > 1$ and equal to it at $\gamma = 1$. A lower bound on u sharpens it further — the extremal law becomes $\{u_{\min}, 1 + \mathrm{Var}/(1 - u_{\min})\}$ — but by less. Against the allowance (`numerics/sharpmom.m`):

γ	1	1.5	2	2.5	3
allowance on A	1.0661	1.0595	1.0561	1.0542	1.0528
A true	1.0411	1.0502	1.0584	1.0671	1.0764
Lyapunov	1.0411	1.0763	1.1215	1.1821	1.2612
Proposition 10.7	1.0411	1.0606	1.0795	1.1003	1.1230
with the lower bound on u	1.0411	1.0585	1.0769	1.0980	1.1213

The verdict is sharper than before, and it is negative. From $\gamma = 2$ the *true* gradient sum exceeds the allowance — 1.0584 against 1.0561 — so no bound on A from any moments can carry the chain: the two-rate estimate itself is too weak there, not the variance step. At $\gamma = 1.5$ the sharp bound misses by 0.001 and the lower-bound refinement passes by 0.001, on the true variance; a certified box maximum of the variance would have to land inside that margin. So the reach of this route is $\gamma \lesssim 1.5$, and the moment question that Section 11 used to list is closed: the bound is the sharp one, and it is not enough.

That is a complete route at logarithmic utility, and a sharp statement of what it needs. It would give Theorem 1 at *every* state, not only the binding one, hence monotone capital supply and uniqueness at $\mu = 1$ — which is more than Proposition 7.4 delivers.

The two-regime bracket

So the variance of the risk tilt is what must be bounded, and bounding it needs an admissible set for C tight enough that its worst member satisfies the requirement. Read as a bound, the constrained regime's identity supplies a *ceiling*: consumption never exceeds cash on hand, so at every state

$$\underbrace{\kappa_k\big(m + H_k^{\mathrm{risk}}(z')\big)}_{L(z')} \;\leq\; C(z') \;\leq\; \underbrace{\min\big\{\kappa_k(m + H_k(z')),\; m\big\}}_{U(z')}, \qquad m = y(z') + Rb,$$

both sides primitive, together with the slope constraints $\kappa_k \Delta y \le \Delta C \le \Delta y$ that link the states. Maximising $\mathrm{Var}_w(M/C)$ over that admissible set changes the picture completely. The best feasible profile found gives 0.0406 at the newborn stage — essentially the true variance of 0.0404 — against an allowance of 0.0661, and a search over the set turns up nothing above it. The slope constraints are what do the work: dropping them and maximising over the box alone gives 0.1537.

It is not a certificate, and the gap is worth being exact about. What is established rigorously is an *interval*: the true consumption profile is feasible, so the maximum is at least 0.0411; the polytope sits inside the box, so the maximum is at most 0.1537. The allowance of 0.0661 lies inside that interval. The search says the maximum is at the lower end and the chain closes; the search is not exhaustive, and at two of the three stages tested it returns a value below the true variance, which shows directly that it can miss.

Closing this needs an upper bound on the maximum that uses the weights. The weight-free one is available and useless: since $\mathbb{E}_w[u] = 1$, $\mathrm{Var}_w(u) \le \max u - 1 \le \mathrm{HM}(U)/\min L - 1$, which is 0.97 at the binding stage against an allowance of 0.066.

The coefficient of variation

The certificate wants an upper bound on the maximum, and the coordinates matter for getting one. Put $v_i = 1/C_i$. Put $v_i = 1/C_i$. Then $M = 1/\sum_j w_j v_j$ and $u_i = M v_i$, so

$$\mathrm{Var}_w(u) \;=\; \frac{\sum_i w_i v_i^2}{\left(\sum_i w_i v_i\right)^2} - 1, \tag{CV}$$

the squared coefficient of variation of v under w — exact algebra, with the mean eliminated rather than bounded. The box in C is a box in v, the slope constraints are still linear in C, and (CV) is a ratio of quadratic forms, scale-invariant, of the kind whose maximum over a polytope is a standard finite computation.

Why enumeration is legitimate

In v-coordinates the objective is coordinatewise quasi-convex, and that is what makes enumeration legitimate. Holding all but one coordinate fixed,

$$\frac{\partial}{\partial v_j} \frac{\sum_i w_i v_i^2}{(\sum_i w_i v_i)^2} \;\propto\; v_j B - A, \qquad A = \sum_{i \ne j} w_i v_i^2, \quad B = \sum_{i \ne j} w_i v_i,$$

so the objective is unimodal with a *minimum* in each coordinate: its maximum over any interval is at an endpoint. At a maximiser every coordinate therefore sits at an endpoint of its conditional feasible range.

Concavity is the missing constraint

The bracket and the slope constraints are not by themselves enough: enumerating the admissible set turns up a profile with variance 0.0874, above the allowance of 0.0661 (Appendix A).

The profiles that do the damage zig-zag across earnings states, and they are not consumption functions: consumption is *concave* in cash on hand, by Carroll–Kimball, which this development already proves (`concaveOn_stageConsumption`). Across earnings states at fixed assets, cash on hand moves with y, so concavity says the slopes $\Delta C/\Delta y$ fall with j — five more linear constraints that the bracket was silently discarding.

With them the same two-directional enumeration gives 0.0415 at every stage tested, against allowances of 0.0661, 0.0725 and 0.0969: a margin of 1.59 to 2.34 times.

stage	59	40	20
allowance	0.0661	0.0725	0.0969
enumerated, bracket + slopes	0.0874 ×	0.0702	0.0415
enumerated, + concavity	**0.0415**	**0.0415**	**0.0415**

So the ingredient the chain was missing is a theorem it already has.

The certificate in exact arithmetic

A first run of this certificate in floating point, over a grid of asset values, leaves two gaps: the arithmetic, and the sampling (Appendix A). Both close. The calibration is rebuilt so that every constant is rational: the earnings chain is Rouwenhorst rather than Tauchen, whose transition probabilities are rational functions of $\rho = 9/10$ and whose invariant law is binomial, and the earnings levels are rational, normalised to mean exactly one. At $\gamma = 1$ everything downstream stays rational — $\mathfrak{p} = \beta R_1 = 624/625$, $q = R_1/R_2 = 52/53$, the propensities κ_k by their recursion, arithmetic human wealth by averages and risk-adjusted human wealth by harmonic means — so the allowance is an exact rational number,

$$\frac{\mathfrak{p}}{q\,(1-\kappa_{J-1}R_1)} - 1 \;=\; 0.066143400439\ldots$$

The computation is carried in integers over 10^{12} with every operation rounded *outward*, so each quantity is a true enclosure; and the asset argument is carried as intervals covering its whole range rather than sampled, which at the newborn stage is $b \in [0, 3]$, since a newborn holds nothing and cannot save more than $y_{\max}$.

The branch and bound certifies every node: seven earnings states, the asset range split into four intervals, at most 3385 boxes at the worst of them. Nothing is sampled and nothing is rounded inward.

It survives a much larger asset range than the newborn needs. Rerun over $b \in [0, 12]$ in eight intervals — four times the range any newborn can reach, and beyond what the later stages of the life cycle require — every node is still certified, with at most 16673 boxes. So the asset argument is not where the remaining scope question lies.

Every stage

Running the certificate at each stage in turn — each with its own κ_k, its own arithmetic and risk-adjusted human wealth, its own allowance, and the asset argument as intervals over

$[0, 12]$ in four pieces — certifies all fifty-nine of them. The allowance falls monotonically from 1.168 at $k = 1$ to 0.066143 at the newborn stage, and every node is discarded at every one.

The cost is where the difficulty lives, and it does not plateau. The early stages are nearly free — stage 1 needs a single box, stage 5 needs 539 — and the count then climbs from about 3600 around stage 40 to 14803 at stage 44 and 32733 at stage 59, a tenfold growth over the last twenty stages as the allowance tightens towards the variance. The whole run takes about ten hours.

What is settled

At the rational calibration, at logarithmic utility, over $r \in [4\%, 6\%]$: the variance is certified below what the damping allows, in exact arithmetic, at every stage and over the whole asset range. With the eight formalised lemmas of this section and Section 7, Theorem 1 holds at every state and every age; capital supply rises with the rate; the stationary equilibrium is unique.

What that is, precisely, is a theorem with a computational hypothesis discharged by certificate rather than by proof term — the same standing as the existence bracket of Section 6, and a weaker standing than the lemmas it rests on. The calibration is one calibration: $\beta = 24/25$, $\rho = 9/10$, seven earnings states, sixty periods, $r \in [4\%, 6\%]$, logarithmic utility. Nothing here extends to $\gamma > 1$, where Section 7 shows the certainty-equivalent side of the argument fails from $\gamma \approx 1.5$ on. The log case is closed; the rest is not.

It has since been superseded at $\mu = 1$. Theorem 8.3 proves Theorem 1 at every state and age for $\gamma \leq 1$ at every calibration, by Light's induction on the stage values, with no computation; the certificate remains as the record of the increment route, which is the only one that says anything above $\gamma = 1$, and Proposition 10.7 says how far it can go. The certificate is `numerics/rational/cert.py`, with the calibration in `inputs.py`. Numerics: `cestep.m`, `damp.m`, `nearfar.m`, `lyap.m`, `varbnd.m`, `varmpc.m`, `varmax.m`, `slopestate.m`, `tworeg.m`, `polymax.m`, `bandb.m`, `vertcert.m`, `vertfull.m`, `certify.m`.

What is proved stands on its own: single crossing needs only the increment (I); the increment is controlled by the envelope of Theorem 10.1; its Lipschitz constant is Theorem 10.2 and Theorem 10.3; and the mass weighting is exact in the coordinate where the failure lives, by Theorem 10.4. Numerics: `incbound.m`, `ebar.m`, `vjbound.m`, `vjdiag.m`, `mpclow.m`, `slopecheck.m`.

11 What remains

11.1 Above logarithmic utility

The one thing the programme set out to do and has not done is uniqueness at $\mu \in \{3, 5\}$. The argument of Section 10 does not fail there for want of propagation: the per-stage multiplier $m_k = qA/(\text{þ} + qA\kappa_k R)$ has a product below one over a life at every risk aversion we tested, up to $\gamma = 5$. What gives way is the certainty-equivalent side. The gradient sum A is one plus the variance of the risk tilt at $\gamma = 1$ and only bounded by $(1 + \mathrm{Var})^{(\gamma+1)/(2\gamma)}$ above it, by

Lyapunov, and both the loss in that inequality and the variance itself grow with γ — the variance tenfold between $\gamma = 1$ and $\gamma = 3$. The bound crosses the allowance between $\gamma = 1$ and $\gamma = 1.5$.

So the question at $\mu = 3$ is whether $\mathbb{E}_w[u^{1+1/\gamma}]$ can be bounded better than by Lyapunov, given $\mathbb{E}_w[u] = 1$ and a bound on the variance. That is a question about moments of a positive random variable with known mean, not about households, and it is where we would look next.

11.2 Routes that are closed

Three, each with a reason worth keeping so that none is reopened.

Mean field games. Section 9. Date-by-date Lasry–Lions monotonicity is false here at every risk aversion, and the summed condition that the argument actually uses, taken to its weakest form, collapses to monotone capital supply. What survives is about the life-cycle household: the reduction of the summed pairing to a mixed difference, the duration wedge for Φ, and the exact re-pricing identity.

Stochastic dominance. Light and Weintraub's Theorem 2 reaches uniqueness from ordered distributions, and the obvious weakening of its hypothesis is to ask only for that ordering: first-order stochastic dominance of the cohort distributions. It buys nothing. In wage units, at $r = 4\%$ against 6%, dominance holds to machine precision at $\gamma \leq 3$, where the policy ordering already holds exactly, and fails at precisely the risk aversions where the policy ordering fails — at $\gamma = 5$, eleven of sixty ages have a crossing of the asset CDF, worst gap 5×10^{-2}, and 0.57 conditional on the earnings state. The distributions genuinely cross, and they stop being ordered exactly when the policies do, while aggregate supply stays monotone throughout. The aggregate is the weaker statement and there is no distributional route to it (`fosdcheck.m`).

Two-sided sandwiches. For the two-rate comparison at a state, the width of the sandwich at a given rate *is* the precautionary wedge: arithmetic human wealth gives the ceiling, risk-adjusted gives the floor, and they cannot both be tight. So bounding the two problems separately works only while the consumption response to the rate exceeds that wedge, which at Aiyagari's numbers is $\gamma = 1$ alone (Section 7). Any route to $\mu > 1$ has to bound the *difference* between the two problems, which is what Section 10 does.

11.3 The classical two-period results

Section 2 does the steady state, and for every risk aversion rather than only for logarithmic utility. What remains of the two-period economy is standalone: monotone global convergence, existence under Inada conditions with a multiplicity witness in general, dynamic inefficiency of a steady state with $r < n$ by an explicit Pareto-improving transfer, the golden rule, and the autarkic and monetary steady states of the pure-exchange economy.

11.4 Extensions

An age profile of earnings; a government budget alongside the pension of Section 5; survival risk with accidental bequests as a second fixed point; population growth. Each of these changes the equilibrium object rather than the household, so the existence argument of Section 6 should carry with the bracket recomputed.

11.5 The certificate

The uniqueness result of Section 10 rested on a hypothesis discharged by a computation in exact rational arithmetic rather than by a Lean proof term. At $\mu = 1$ that is no longer needed: Theorem 8.3 proves the same conclusion, at every calibration, by proof term. The certificate keeps its value as the record of the increment route, the only one that reaches above $\gamma = 1$, and the moment question is settled by Proposition 10.7: the sharp bound is $(1 + \mathrm{Var})^{1/\gamma}$, and from $\gamma = 2$ the true gradient sum already exceeds the allowance, so the route stops at $\gamma \approx 1.5$ whatever bound is used. Replaying the certificate inside Lean would now only certify a case already proved.

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A Routes that do not work, with their numbers

Four of the arguments in this paper reached their form by elimination, and the eliminated routes are worth recording with the numbers that killed them: each is a natural thing to try, and each fails for a reason that says something about the model. They are gathered here, in the order of the sections they belong to — the existence bracket of Section 6, Theorem 1 in Section 7, the mean field route of Section 9, and the increment argument of Section 10, where the allowance to be met is 0.0661 at the newborn stage.

Where the existence bounds stop

The bracket of Section 6 is not tight at either end, and the same fact stops it at both.

Where the bounds stop

Both ends are now within a few points of the limit, and neither will move further with the tools here. The reasons are worth recording, because they are the same reason twice.

At the top, the floor's entire loss is in one inequality. Running the floor's own recursion on the cohort's *true* mean consumption path reproduces aggregate capital exactly, 6.180 against 6.180 at $r = 8\%$ (`numerics/diag.m`), so none of it is in the aggregation, the affine bound or the earnings weights: all of it is in $c_k \le \kappa_k(m + H_k(z))$. That inequality is exact at the end of life and a fifth loose at the start, and the leverage is what hurts. At birth the household earns 1 and consumes 0.698, so it saves 0.302; the bound allows 0.858 and leaves a provable saving of 0.142. Overstating consumption by a fifth halves the saving.

Three ways of tightening it fail. Truncating the horizon gives $c_k \le \kappa_j(m + H_j(z))$ for every $j \le k$, all valid, but $\kappa_j(m + H_j)$ is decreasing in j so the minimum is the full-horizon branch already in use. Substituting the power mean that the Euler step actually produces, in place of the arithmetic mean the proof relaxes it to, is not legitimate: the step shifts by gross saving, and a power mean with negative exponent is superadditive, so the shift comes out larger inside than outside, which is the wrong way for an upper bound; carried through anyway it violates the true consumption function by 0.30 and gives a floor of 7.45 against a true supply of 6.18. Putting the shift inside the mean avoids superadditivity but forces tomorrow's bound to be read at a saving ceiling, decoupling the budget identity from the Euler step, and loses more than the power mean gains, 0.956 against 0.858 at birth. That last one also measures the sensitivity: with the *true* saving the same recursion reproduces the truth exactly, and the ceiling it must use instead is only 1.19 times the truth at birth.

And the same obstruction from the other side

At the bottom there is no structural slack left. Reading the ceiling at the assets held, with the same human wealth, gives the identical number at every rate, 4.8147 against 4.8147 at $r = 2\%$ (`numerics/ceilat.m`): given its one-step bound the affine ceiling is already exact. What remains is the one-step bound, and there the asset-dependent incidence bound of Section 7.16 does help, because the affine ceiling averages over states and so picks up the low earnings states where the cap bites. It is worth 0.7 points, the same as the clamp gave at the top: the low endpoint would move from 2.78% to 3.47%.

The proof does not follow. Taking those points needs the ceiling read at the assets held, which needs its argument, the saving ceiling, to increase with assets. It does, with not one decreasing step over the grid, and the truth sits exactly on the bound the induction would need: the largest slope of the asset-dependent human wealth is 26.147 and the cap's slope is 26.147, because it is the cap branch that binds. But carrying that bound through the recursion means passing a perturbation through the power mean, and a power mean with negative exponent is not 1-Lipschitz in the sup norm. It is superadditive, so a uniform shift comes out amplified, here by 9.1; over fifty-nine ages the natural induction would allow the slope to grow by 4×10^{56}, against a truth that does not grow at all (`numerics/slopes.m`). The cheap way round, reading the asset-dependent human wealth at zero assets, is valid and

monotone by construction and moves the endpoint from 2.78% to 2.81%.

So superadditivity of the power mean blocks both ends: at the top it stops the mean being substituted into the consumption upper bound, and at the bottom it stops a slope bound being propagated past it. It is the same obstruction that closed the rate-response bound in the stationary companion.

Theorem 1: the routes that do not close

Why the natural induction does not close

The backward induction that the transfer lemma suggests compares the two Euler equations at a state where the lower-rate household saves. Writing $b = g^{(1)}_{k+1}(a, z)$ for the pivot, the induction hypothesis and monotonicity of c_k give $c^{(2)}_k(b') \le c^{(1)}_k(b) + \Delta r\, b$ at the relevant points, and the step closes provided

$$R_2\, \mathbb{E}\big[(c^{(1)}_k(b) + \Delta r\, b)^{-\mu}\big] \;\ge\; R_1\, \mathbb{E}\big[c^{(1)}_k(b)^{-\mu}\big]\Big(1 + \frac{\Delta r\, a}{c^{(1)}_{k+1}(a, z)}\Big)^{-\mu}.$$

To first order this asks $\mu\big(b/c^{(1)}_k(b) - a/c^{(1)}_{k+1}(a, z)\big) \le 1/R$: the ratio of assets to consumption may rise from one age to the next by at most $1/(\mu R)$. It does not. At zero wealth in the highest income state the left-hand side is large, because such a household saves a great deal against a certain fall in income, while the right-hand side is about one. The margin is negative from the third stage of life onward for every degree of risk aversion we tried, logarithmic utility included, at -0.12 for $\mu = 1$, -2.86 for $\mu = 3$ and -6.29 for $\mu = 5$ per unit of the rate gap. Keeping the term the step discards, the marginal propensity to consume times the size of the hypothetical violation, bounds the violation by a multiple of Δr rather than eliminating it, which is not enough: the aggregate difference the equilibrium argument must sign is itself of order Δr.

The risk half, and why it resists

Two caveats remain. The constant 0.36 is measured over fifteen calibrations and is not proved. And both are statements at zero wealth in the highest earnings state, which is where the life-cycle model binds; the reduction of Theorem 7.1 is what would carry them to the rest of the state space, and its own condition fails, so this is a sharp description of where the truth lies rather than a proof of it.

It is worth being precise about why the constant resists. What has to be bounded is a difference of *rate responses* between two problems, the risky one and its certainty equivalent, and the threshold is the rate at which one of those responses changes sign. No bound on the *level* of consumption helps with that, however sharp: at Aiyagari's calibration the sharpest sandwich available has width 0.30 to 0.48 at the state in question, while the response itself runs from -2.48 at $r = 2\%$ through zero to $+0.29$ at $r = 6\%$ (`numerics/scale.m`). A bound on where consumption lies says nothing about how it moves. What is needed is a bound on the rate response itself, and that is exactly the object the companion write-up proved cannot

be had from the affine induction, whose per-step multiplier is at least $1/\text{þ} > 1$. The two halves of (2) therefore stand on different ground: the first is now a theorem for every risk aversion, and the second is blocked by the same obstruction that closed the rate-response bound.

Theorem 1 with a margin. Light's invariant fails above $\gamma = 1$ inside the reachable region (`gapreach.m`: at $\gamma = 2$ from $a = 2.8$ with one period to go, where a cohort can reach 7.6), because κ_k rises with the rate when $\gamma > 1$, so an old household with modest wealth consumes proportionally more at the higher rate than $\rho^{1/\gamma}$ allows — while Theorem 1 itself holds there on the interest-income term. So the induction needs a different invariant, and the natural one is Theorem 1 with a margin, $g_k(a; R_2) - g_k(a; R_1) \geq s_k(a, z) \geq 0$, fed through the Euler step with the propensity floor. It does not close. With the margin set to a fraction ϕ of the *true* stage difference and the consumption slope over $[b_1, b_1 + s]$ taken exactly — the least lossy version of the chain — the step fails at 22% of reachable states at $\gamma = 2$ and 36% at $\gamma = 3$ for $\phi = 0$ (this is the slack condition), still at 19% and 32% for $\phi = 0.75$, and only at $\phi = 1$ does it reduce to the 1% of corner states where the higher-rate household's Euler equation is an inequality; with the unit consumption slope the margin makes things worse at every ϕ (`twosided.m`, `twosided2.m`). The chain carries the margin one-for-one from tomorrow's marginal utility to today's, so a margin helps only where it exceeds what today demands, and a primitive margin would have to be within about a quarter of the truth at every state. The one natural primitive candidate, the certainty-equivalent household's extra saving, is off by a factor between 1.1 and 6.7 across states (median 1.5 at $\gamma = 2$, 2.2 at $\gamma = 3$) and fails at two thirds of states (`twosided3.m`). Closed.

The borrowing-constraint decomposition. Section 7 records that loosening the limit moves the threshold in risk aversion from 3.42 to 4.38, and it was natural to hope for Theorem 1 as (loose-limit response) + (constraint cross term) with the first half easy. Measured across the reachable state space (`nbl.m`: a stage-dependent limit at 0.9 and 0.5 of the natural one, itself rate-dependent), the loose-limit household *violates* Theorem 1 at 31% of its non-negative-asset states at $\gamma = 2$ and 52% at $\gamma = 3$, by up to 0.58 in saving units; the cross term is non-negative at every state, with median 0.5 and ninetieth percentile 3.9 times the loose response. The threshold statement is about one state, the newborn with nothing in the top earnings state; in the bulk of the state space the constraint is what carries Theorem 1, and the decomposition adds difficulty rather than removing it. Its slack condition is no easier either (11–21% of states fail). Closed.

The mean field condition, measured

Section 9 rests on two measurements of the Lasry–Lions pairing and two more of the condition (W) it reduces to. They are here.

Date by date it fails, and it fails at birth

Take two economies differing only in β, each at its own equilibrium, so that each distribution is priced by the aggregate it actually carries. Since β does not appear in F_t, this is a legitimate test of the date-t condition. At the calibration of Section 6 ($\alpha = 9/25$, $\delta = 2/25$, $J = 60$, Aiyagari's earnings chain), with $\beta = 0.960$ against $\beta = 0.950$:

γ	0.5	1	2	3	5
dates with $P_t < 0$ (of 60)	3	4	5	6	6
P_0	−0.00147	−0.00227	−0.00324	−0.00419	−0.00804
$\sum_t P_t$ (undiscounted)	+0.271	+0.175	+0.091	+0.061	+0.047

The failures are at the start of life, and they survive as the two equilibria close up: $P_0/(\Delta\beta)^2$ converges to -25.6 at $\gamma = 1$ and -44.4 at $\gamma = 3$ as $\Delta\beta$ falls from 0.02 to 0.00125, with the count of negative dates unchanged throughout. So this is a property of the derivative, not an artefact of a wide gap.

Date zero is the sharp case, and it is worth spelling out, because it is exactly the common initial distribution that was supposed to rescue the argument. Every cohort is born with nothing, so $a_0 = 0$ for all of the mass in both economies, and the affine-utility pairing above is identically zero: with no assets to price, the aggregate coupling has no grip at all. But the wage still differs, and the utility gain from the higher wage is weighted by marginal utility, which is higher in the economy that saves more. At $\gamma = 1$ mean date-zero saving is 0.2937 against 0.2736, and P_0 is strictly negative. The common start does not make the date-zero pairing vanish; it removes the one term that could have made it positive.

Summed, it holds where the equilibrium lives

The summed condition is a different matter. Test it over the family the equilibria belong to: $m(\hat{K})$ generated by optimal life-cycle behaviour at the prices of $\hat{K}$, priced inside F by the aggregate $A(\hat{K})$ it actually carries. Every equilibrium is an $m(\hat{K})$ with $A(\hat{K}) = \hat{K}$, so monotonicity over pairs from this family is exactly what Theorem 9.2 needs.

γ	1	2	3	5
pairs with $\sum_t \beta^t P_t < 0$, $\hat{K}/K^* \in [0.5, 2]$	1/66	0/66	0/66	0/66
pairs failing inside $K \in [3.55, 7.40]$	0/36	0/36	0/36	0/28
least margin $\sum_t \beta^t P_t/(A_1 - A_2)^2$ there	+0.075	+0.097	+0.121	+0.338

The bracket $K \in [3.55, 7.40]$ is the one $r \in [2\%, 8\%]$ of Theorem 6.18, inside which the equilibrium is proved to lie. Within it the summed pairing is strictly positive for every pair and every risk aversion tested, with a margin bounded away from zero, and the margin grows with γ. The single failure is at $\gamma = 1$ between two candidates one of which sits outside the bracket entirely.

A pointwise condition

Submodularity of the flow $u\big(w(K)y + (1+r(K))a - a'\big)$ in (K, a) is, for constant relative risk aversion, exactly

$$\gamma\,(1+r(K))\,(a_t - Ky_t) \;\le\; c_t \qquad \text{at every age and state,} \tag{P}$$

which implies (W) and hence the summed condition. It is free wherever the household holds less than its share of the capital stock, and it is hopeless where it does not: at the calibration it is violated on 27% to 45% of the mass, at every risk aversion. Requiring each date to carry itself is precisely what the summed condition was introduced to avoid, and (P) is that mistake made again one level down.

Where the pointwise ordering fails

(W) is a single scalar comparison, and it is weaker than the pointwise ordering of the two plans' asset paths. Measured at the calibration, on plans drawn from the candidate family at $\hat{K}/K^* = 0.90$ against 1.10 — so the two plans face different wages as well as different rates, which is the comparison (W) itself involves:

γ	1	2	3	5
$\Phi_1 - \Phi_2$ at the midpoint K	+39.36	+12.67	+7.34	+7.32
mass on which $a_t^1 \ge a_t^2$ FAILS	0.0000	0.0025	0.0409	0.2006
the same, at a common wage	0	0	0	0.0091

The second row is the one to read against Light's Theorem 1, which is a statement about the household at a *fixed* wage: in wage units the policy ordering is exact — violated on zero mass, not on small mass — for $\gamma \le 3$, and fails on under one percent of the mass at $\gamma = 5$ (on 6.7% at $\gamma = 8$). The first row, which lets the wage move along the frontier as well, overstates the failure of Theorem 1 by more than an order of magnitude; it is the right measure for (W) and the wrong one for Theorem 1. So (W) is weaker than the pointwise ordering and does survive where that ordering fails, but the margin between them at $\gamma = 5$ is one percent of the mass, not a fifth.

More than those pairs is true. Sweeping the planning aggregate over $\hat{K}/K^* \in [0.5, 2]$ at fourteen points, $\Phi(K, p(\hat{K}))$ is strictly *decreasing* in $\hat{K}$ at every one of the thirteen steps, for $\gamma = 1, 3, 5$ and for each of $K = 0.90K^*, K^*, 1.10K^*$. So (W) holds not merely at the pairs one happens to test but as a monotone comparative static in the plan. Over all 91 pairs at five pricing points each, $\gamma = 3$ and $\gamma = 5$ give no failures at all; the 42 failures out of 455 at $\gamma = 1$ all involve the extreme candidate $\hat{K} = 0.5K^*$, whose carried aggregate is 48.6, ten times the capital stock and far outside the bracket of Theorem 6.18.

The increment route

Where the failure lives

Measured at $\gamma = 5$ over $r \in [4\%, 6\%]$, the failure reaches down to $a = 0$ — so no threshold in assets confines it, and a Markov bound on a thin upper tail is worthless: the sup of the failure below any threshold is the global sup, and the resulting bound is the uniform one that already failed. But sorted by earnings state the picture is different. With Aiyagari's seven-state chain the sup of the failure over assets, state by state, is

$$(0,\ 0,\ 0,\ 0,\ 0,\ 0.0143,\ 0.0460),$$

zero in the bottom five states and confined to the top two, whose invariant mass is $\nu_6 + \nu_7 = 0.095$. At $\gamma = 8$ it reaches the fourth state and the profile is $(0, 0, 0, 0.0014, 0.0118, 0.0333, 0.0701)$. The failure of Theorem 1 is an *earnings-state* phenomenon: it is the high-earners, who are few.

The minimum-based step amplifies

The Theorem-1 step can be made quantitative without touching a power mean. Suppose the failure $D = c_2 - c_1 - \Delta r\, a$ is positive at (a, z). Then $b_1 > b_2 \geq 0$, so the low-rate Euler equation holds above the floor and the high-rate one under the cap; the stage-k failure bound and monotonicity of consumption in assets give $C_2(z', b_2) \leq C_1(z', b_1) + \Delta r\, b_1 + \varepsilon_k(z')$, and $(C + s)^{-\gamma} = C^{-\gamma}(1 + s/C)^{-\gamma} \geq C^{-\gamma}(1 + \sigma)^{-\gamma}$ with $\sigma = \max_{z'} s(z')/C_1(z', b_1)$. Chaining the two Euler relations,

$$D \ \leq\ c_1\Big[(R_1/R_2)^{1/\gamma}(1 + \sigma) - 1\Big] - \Delta r\, a. \tag{Q}$$

This is a genuine quantitative step: it recovers the no-failure criterion when the bracket is nonpositive, and bounds the failure otherwise. It also amplifies. The derivative of (Q) in ε_k is $c_1(R_1/R_2)^{1/\gamma}/\min_{z'} C_1(z', b_1)$, measured at 4.7 at $\gamma = 5$ and 4.2 at $\gamma = 8$. Over sixty stages the recursion started from the exact terminal condition $\varepsilon_0 = 0$ reaches $V \approx 2.8 \times 10^3$ at $\gamma = 5$ against a true supremum of 0.046 — and with the sandwich floor in place of the true C_1, 10^{60}. This is the amplification the rate-response route already ran into, with a per-step multiplier bounded below by $1/\text{þ} > 1$; the life cycle does not escape it, it merely makes it finite.

The direct bound is too loose

The alternative is to bound the failure at each stage separately, with no induction, from the sandwich: $c_2 \leq \kappa_k^2(m_2 + H_k^2)$ and $c_1 \geq \kappa_k^1 m_1$. That gives 1.835 against an allowance of 0.032, too loose by a factor of 58, and the source of the looseness is visible: the sandwich slack is of order $\kappa_k H_k$, measured at 0.96, an order-one quantity where the target is of order 0.03.

What the two failures have in common

The remaining gap is not a matter of assembling existing bounds. It needs an estimate of the rate-response of consumption whose error is small *relative to the response itself* — an absolute

error below 0.03 on a quantity of size about one — and no sandwich in this programme is of that quality. That is a sharper statement of the requirement than the programme has had before, and it is attached to a single distribution-free object rather than to Theorem 1 in full.

Bounding the multiplier

A has to be bounded from primitives, and the obvious bound destroys the gain. Termwise, $\sum w\,x^{-\gamma-1} \le \sum w\,x^{-\gamma}/\min x$, so $A \le M/\min x$ (`sum_mul_rpow_pred_le`) — and $M/\min x$ is precisely the multiplier that bounding the expectation by its smallest term gives. Measured, A runs from 1.04 to 1.19 while $M/\min x$ runs from 4.7 to 5.9; substituting the bound raises m_k to 2.4–2.6 and the product to 10^{20}. The crude route returns exactly to where it started.

So A's smallness is a statement about the *weights* of the earnings chain and not its support: with $\rho = 0.9$ the transition puts almost all mass on neighbouring states, where next period's consumption is close to its certainty equivalent, while the far states that make $\min x$ small carry almost no probability.

Near states and far states

The obvious way to use the weights is to split the states, bounding the near ones by proximity and the far ones by their probability. It fails, and instructively. With $d = 1$ the far mass is 3×10^{-3}, so the far term is negligible — and the bound is still 2.48 against an allowance of 1.066, because the near term takes a *maximum* over neighbouring states and $(C/M)^{-\gamma-1}$ reaches 2.5 even there. Widening to $d = 2$ makes it worse, at 4.99. The averaging is the whole of the effect, and any bound that replaces it by a maximum on a subset throws it away.

Why level brackets cannot bound a variance

Two ways to bound $\mathrm{Var}_w(u) \le 0.066$ were tried. Both fail, and the second fails for a reason worth stating.

Bounding it through *levels* — $\mathbb{E}[u^2] = M^2 \sum w/C^2$ with the ceiling on M and a floor on C — gives 5.5 with the constraint-incidence floor and 11.0 with the lower sandwich, against an allowance of 0.066: too loose by factors of 84 and 166. The reason is the same one that closed the two-sandwich route at $\gamma = 2$, one level down. A variance is a measure of *spread*, and a per-state bracket of width comparable to the spread contributes a spurious spread larger than the real one; squaring makes it worse.

Bounding the spread directly is the right instinct. Across income states at fixed assets, cash on hand differs by $y(z') - y(z'')$, so the propensity brackets the consumption difference: $\kappa_k|\Delta y| \le |\Delta C| \le s|\Delta y|$ with s any proved upper bound on the propensity. With the trivial $s = 1$ and the incidence floor anchoring the level, maximising $\mathrm{Var}_w(M/C)$ over the admissible set — $C(1)$ in its bracket, each increment at either end — gives 0.393 at the newborn stage, against the allowance of 0.066. Short by a factor of six. (A single admissible profile gives 0.044 and appears to pass; it is not the worst case, and the worst case is what a bound must survive.)

Scaling suggests the slope bound needed is about $s \leq 0.41$. That cannot be uniform: the true cross-state slope is $\kappa_k \approx 0.04$ where the household is unconstrained but exactly 1 where the borrowing constraint binds.

The slope, state by state

The split is available. The saving floor of `le_stagePolicy_state_on` certifies that the constraint cannot bind at (b, z') whenever $(1 - \kappa_k)(y(z') + Rb) > \kappa_k H_k(z')$, a primitive test, and it certifies 89% to 94% of the pairs that arise. Only 1.6% are in fact constrained.

What it does not do is let the constrained states be neglected. Measured, there are pairs (z, b) at which the states carrying the entire transition mass are constrained — low assets and low earnings, where the household consumes its cash on hand — so the weights cannot be used to make that regime disappear.

The way through is not a better slope bound on that regime but a different handle on it. Where the constraint binds, consumption is cash on hand *exactly*: $C(z') = y(z') + Rb$, with no bracket at all, so the spread of C across those states is exactly the spread of y, which is known. The two regimes want treating differently in kind — an identity on the constrained set, a slope bound on the unconstrained one — rather than a single slope bound made state-dependent. The worst-case worst-case maximisation over the bracket is loose precisely because it gives the constrained states a bracket they do not have.

Branch and bound with an inconsistent bound

The natural certificate is branch and bound: on a sub-box $[c^-, c^+]$ the harmonic mean is increasing in each coordinate, so $M \leq M(c^+)$ and $u_i \leq M(c^+)/c_i^-$, giving an upper bound on $\mathrm{Var}_w(u)$; branch, prune the slope-infeasible boxes, and refine. It does not converge: with four thousand nodes per state the certified bound is 4.34 against an allowance of 0.066, and the worst node is the lowest earnings state, where the bracket is widest.

The bound is the problem, not the budget. It evaluates the mean at the top of the box and the individual consumptions at the bottom, which no single point achieves — an inconsistency that no amount of branching removes, because it is reintroduced at every level.

Incomplete searches, and the profile they miss

Enumerating those endpoints settles the question, and not in favour of the bracket. A forward sweep over the chain — each coordinate at an endpoint of the range its predecessor leaves — gives 0.0414, and random probes never beat it, which looks like closure. Enumerating patterns in both directions, so that a coordinate may also be pinned against its successor, gives **0.0874** at the newborn stage, against an allowance of 0.0661. The admissible set provably contains a profile that violates the requirement. The forward sweep was incomplete, and so were the coordinate ascent and the random search that preceded it, all three by a factor of two.

A first certificate, in floating point

The upper bound follows from the same unimodality that justified enumerating. On *any* box the objective is maximised at a box vertex, so the maximum over a box is *exactly* the maximum over its 2^7 vertices — an exact and consistent bound, unlike the one that defeated branch and bound before, which put the mean at the top of the box and the consumptions at the bottom. Branch and bound with it: discard a box when it is disjoint from the polytope, by an interval test on the slope and concavity constraints, or when its exact maximum is at most the allowance; otherwise split the widest coordinate. If the stack empties, every point of the polytope lay in a discarded box and the maximum is certified below the allowance.

It empties. At the newborn stage, over forty asset values in each of the seven earnings states, every node is discarded, with at most 6971 boxes examined at the worst of them.

The newborn stage is the one that matters. The allowance is $\text{þ}/(q(1-\kappa_k R)) - 1$ and κ_k rises as the horizon shortens, so the allowance rises too — 0.0661 at $k = 59$, 0.0725 at $k = 40$, 0.0969 at $k = 20$, 0.2586 at $k = 5$ — while the variance stays at 0.0415 throughout. Certifying the youngest cohort certifies every age, and that step is an argument rather than a sample.

Its two gaps

At the calibration, at logarithmic utility, over $r \in [4\%, 6\%]$: every input of the chain holds. The error carried between stages is damped (Theorem 10.2, `damped_le_one`), the gradient sum is one plus a variance (Proposition 10.6), and the variance is certified below what the damping allows. Theorem 1 then holds at *every* state, hence capital supply rises with the rate, hence the stationary equilibrium is unique.

Bounding the failure from primitives. For $\gamma \leq 3.4$ the mass-weighted failure is exactly zero and the budget over $[4\%, 6\%]$ is 0.670, so the increment route needs only a *provable* per-age failure bound of order 0.01, never Theorem 1 itself. The failure at a state is $c_2 - c_1 - \Delta r\, a$, so it is bounded by a ceiling on c_2 less a floor on c_1 (`incprim.m`). With the asset-incidence floor at R_1 and the true c_2, the accumulated bound is 0.17 at $\gamma = 1.5$, 0.56 at $\gamma = 2$ (a pass, by 1.2×) and 1.33 at $\gamma = 3$: the floor side is nearly good enough. With the arithmetic human-wealth ceiling at R_2 and the true c_1 it is 1.53, 2.57 and 3.96: the ceiling is where the loss is, and it fails even at $\gamma = 1$ (0.74). The worst states are at zero wealth in the lowest earnings state, where the certainty-equivalent ceiling ignores precaution. A power-mean ceiling that keeps the Euler step exact but evaluates tomorrow's ceiling at the saving ceiling implied by the consumption floor is worse still — 1.18 times the truth at the first stage and 1.35 by the thirtieth at $\gamma = 1$, 1.45 and 1.59 at $\gamma = 3$ — because the argument error compounds through the stages. The exact version of that recursion is the Coleman operator itself, whose iterates from the terminal ceiling are the true policy. So no primitive ceiling of the kinds available is within the factor of ten the route needs, and what remains of the route is a rigorous enclosure of the two policies — interval arithmetic on the endogenous-grid recursion, with monotonicity and concavity of consumption (both proved) closing the gaps

between grid points — fed into the increment envelope. That is a certificate of the same standing the logarithmic case had before Theorem 8.3, at $\mu = 3$ where no analytic route is known.

B Index of formal results

All in the namespace `LeanEconomics.IncomeFluctuation`, directory `LeanEconomics/OLG`. Every result depends only on the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) and none uses `sorry`.

Result	Lean name	Module
Stage value, policy, consumption	`stageValue`, `stagePolicy`, `stageConsumption`	OLG/LifeCycle
Last period: no saving	`stagePolicy_zero`	OLG/LifeCycle
Last period: consume cash on hand	`stageConsumption_zero`	OLG/LifeCycle
Positive consumption at every stage	`stageConsumption_pos`	OLG/LifeCycle
Saving and consumption rise with wealth	`stagePolicy_mono`, `stageConsumption_mono`	OLG/LifeCycle
Carroll–Kimball at every stage	`concaveOn_stageConsumption`	OLG/LifeCycle
Euler under the cap, by stage	`stage_euler_le`	OLG/LifeCycle
Euler above the floor, by stage	`stage_euler_ge`	OLG/LifeCycle
Finite-horizon propensities, human wealth	`stageMPC`, `stageHumanWealth`	OLG/LifeCycle
Propensities in $(0, 1]$, above κ	`stageMPC_pos`, `stageMPC_le_one`, `minMPC_le_stageMPC`	OLG/LifeCycle
Lower Euler step	`crra_stageMPC_step`	OLG/LifeCycle
Upper Euler step	`crra_stageUpper_step`	OLG/LifeCycle
Theorem 3.1, lower	`stageMPC_mul_le_stageConsumption`	OLG/LifeCycle
Theorem 3.1, upper	`stageConsumption_le_stageHumanWealth`	OLG/LifeCycle
One-step saving bound	`stagePolicy_le`	OLG/LifeCycle
Retired household (lump-sum pension)	`withPension`	OLG/Retirement
Cash on hand rises with assets	`withPension_resources_strictMono`	OLG/Retirement
Human wealth is state-independent	`stageHumanWealth_withPension_const`	OLG/Retirement
Perpetuity bound $H_k \le p/r$	`stageHumanWealth_withPension_le`	OLG/Retirement
Life-cycle value with retirement	`lifeValue`	OLG/Retirement
Saving, consumption rise with wealth	`lifePolicy_mono`, `lifeConsumption_mono`	OLG/Retirement
Proposition 2.1	`crra_le_tangent`	OLG/Diamond
Closed-form saving	`saveShare`, `saveShare_euler`	OLG/Diamond
It is the optimum	`saveShare_isOptimal`	OLG/Diamond
Proposition 2.2	`saveShare_strictMonoOn`, `saveShare_strictAntiOn`	OLG/Diamond
Proposition 2.3	`div_one_add_rpow_strictMono`	OLG/Diamond
Theorem 2.4	`exists_unique_steadyState`	OLG/Diamond
Theorem 4.1	`stageConsumption_eq_of_interior`	OLG/Deterministic

Result	Lean name	Module
Rate-free propensity at log	`stageMPC_one_eq_of_rate`	OLG/ Deterministic
Human wealth falls with the rate	`stageHumanWealth_antitone_rate`	OLG/ Deterministic
Theorem 4.2	`stagePolicy_mono_det_log`, `olgCapital_mono_det_log`	OLG/ Deterministic
Chord of a concave power	`CertaintyEquivalent.chord_le_rpow`	OLG/Certainty Equivalent
Theorem 7.3	`CertaintyEquivalent.consumption_antitone_of_chord`	OLG/Certainty Equivalent
Pension schedules	`lumpSum`, `tapered`, `cliff`	OLG/MeansTest
Cash on hand	`cash`	OLG/MeansTest
Theorem 5.1	`isOptimalSaving_mono`	OLG/MeansTest
Monotone in assets when cash is	`isOptimalSaving_mono_assets`	OLG/MeansTest
Gentle taper keeps cash rising	`cash_strictMono_tapered`	OLG/MeansTest
A cliff reverses cash on hand	`cash_lt_cash_of_cliff`	OLG/MeansTest
Theorem 5.2	`exists_consumption_decrease`	OLG/MeansTest
Theorem 5.3, cliff	`exists_saving_decrease_in_assets`	OLG/MeansTest
Theorem 5.4	`policyOf_mono_of_lt`	OLG/MeansTest
One age's expectation operator	`stageStep`	OLG/ Equilibrium
Monotone class preserved	`monoRegion_stageStep`	OLG/ Equilibrium
Ordered policies, ordered functions	`stageStep_le_stageStep`	OLG/ Equilibrium
Life-cycle assets, aggregate capital	`cohortAssets`, `olgCapital`	OLG/ Equilibrium
Theorem 6.1	`cohortAssets_le`, `olgCapital_le_of_stagePolicy_le`	OLG/ Equilibrium
Proposition 6.9	`le_stagePolicy`	OLG/ Equilibrium
Theorem 6.2	`olgEquilibriumRate_unique`	OLG/ Equilibrium
Argmax at a concave continuation	`argmax_eq_singleton_of_concaveSlices`	OLG/Continuity
Slice commutes with the operator	`sliceAt_bellman`, `sliceAt_augStageValue`	OLG/Continuity
Augmented stage policy	`augStagePolicy`, `augStagePolicy_eq`	OLG/Continuity
Jointly continuous stage policy	`continuousOn_augStagePolicy`	OLG/Continuity
Cohort recursion in the augmented state	`augCohortAssets`, `augCohortAssets_eq`	OLG/Continuity
Theorem 6.3	`continuousOn_olgSupply`	OLG/Continuity
Theorem 6.4	`exists_olgEquilibrium`	OLG/Continuity
Saving rises iff consumption responds little	`stagePolicy_le_iff_consumption_le`	OLG/ RateMonotone
Terminal stage, unconditional	`stagePolicy_mono_zero`	OLG/ RateMonotone

Result	Lean name	Module
Condition (1)	`StageSlack`	OLG/ RateMonotone
Theorem 7.1, step	`stagePolicy_mono_step`	OLG/ RateMonotone
Theorem 7.1, induction	`stagePolicy_mono_of_stageSlack`	OLG/ RateMonotone
Theorem 7.1, aggregate	`olgCapital_le_of_stageSlack`	OLG/ RateMonotone
Human wealth, price of the path	`humanWealth`, `pathPrice`	OLG/Certainty Equivalent
Closed form $c_0 = W/D$	`consumption_mul_pathPrice`	OLG/Certainty Equivalent
Saving rises iff the cross product falls	`consumption_antitone_iff`	OLG/Certainty Equivalent
Theorem 7.2	`consumption_antitone_of_le_one`	OLG/Certainty Equivalent
Same household, different limit	`withFloor`	OLG/Borrowing Limit
Stages at a general limit	`finiteValue`, `finitePolicy`, `finiteConsumption`	OLG/Borrowing Limit
Theorem 7.5, consumption	`finiteConsumption_le_of_floor_le`	OLG/Borrowing Limit
Theorem 7.5, saving	`le_finitePolicy_of_floor_le`	OLG/Borrowing Limit
Theorem 7.6	`finiteConsumption_le_shift`	OLG/Borrowing Limit
Terminal slope is R	`stageConsumption_sub_zero`	OLG/ MPCBound
Theorem 7.7	`stageConsumption_sub_le_of_slope`	OLG/ MPCBound
Guaranteed human wealth	`minHumanWealth`, `lowMPC`	OLG/Lower Sandwich
The corner is respected	`lowMPC_mul_minHumanWealth_le`	OLG/Lower Sandwich
Theorem 7.8	`minHumanWealth_le_stageConsumption`	OLG/Lower Sandwich
Power mean under a shift	`powerMean_add_const_le`	OLG/Incidence
Risk-adjusted, capped human wealth	`riskHumanWealth`	OLG/Incidence
Theorem 7.9	`riskHumanWealth_le_stageConsumption`	OLG/Incidence
Theorem 7.10	`cap_necessary_of_corner`	OLG/Incidence
Definition 6.5	`StageRegions`, `allRegions`	OLG/LifeCycle
Sandwich on a reachable family	`stageConsumption_le_stageHumanWealth_on`	OLG/LifeCycle
Floor on a reachable family	`le_stagePolicy_on`	OLG/ Equilibrium
Crude reachable bound	`reach`, `reach_mono`	OLG/Reachable

Result	Lean name	Module
Proposition 6.6	`reachRegions`	OLG/Reachable
Sandwich along a cohort's path	`stageConsumption_ le_ stageHumanWealth_reach`	OLG/Reachable
Floor along a cohort's path	`le_stagePolicy_reach`	OLG/ Equilibrium
Floor at the household's own state	`le_stagePolicy_state_on`	OLG/ Equilibrium
Proposition 6.13	`olgCapital_le_of_invariant`	OLG/Existence
Affine coefficients	`cohortSlope`, `cohortFloor`	OLG/Existence
Proposition 6.10	`cohortFloor_ add_ mul_ le_ cohortAssets`, `le_olgCapital`	OLG/Existence
Mean human wealth recursion	`sum_mul_stageHumanWealth_succ`	OLG/Existence
Theorem 6.11	`stageMPC_ mul_ sum_ stageHumanWealth_le`	OLG/Existence
The aggregate floor climbs	`sum_mul_cohortFloor_mono`, `sum_mul_ cohortFloor_nonneg`	OLG/Existence
Theorem 6.16	`exists_ olgEquilibrium_ of_ bounds`, `exists_olgEquilibrium_rate`	OLG/Existence
Proposition 6.7	`stageMPC_mul_mpcSum`, `stageMPC_mul_ one_sub_pow`	OLG/LifeCycle
The geometric sum is bounded	`sub_mul_mpcSum_le_one`	OLG/LifeCycle
Constant-coefficient slope	`cohortSlope_mul_mpcSum_succ`	OLG/Existence
Proposition 6.8	`mul_geomSum_le_cohortSlope`	OLG/Existence
Human wealth in closed form	`sum_mul_stageHumanWealth_eq`	OLG/Existence
The floor in closed form	`geomFloor_le_sum_mul_cohortFloor`	OLG/Existence
Theorem 6.19	`exists_ olgEquilibrium_ of_ geomBounds`	OLG/Existence
Definition 8.1	`ValueGap`	OLG/Rate Monotone
Theorem 8.2	`stagePolicy_ mono_ of_ valueGap`, `stagePolicy_mono_of_valueGap_all`	OLG/Rate Monotone
Capital from the value gap	`olgCapital_le_of_valueGap`	OLG/Rate Monotone
Theorem 8.3, stage induction	`stageValue_ incDiffSlices`, `valueGap_zero_of_light`	OLG/Light Induction
Theorem 8.3, Theorem 1	`stagePolicy_ mono_ light`, `stagePolicy_mono_light_crra`	OLG/Light Induction
Theorem 8.3, capital and uniqueness	`olgCapital_ le_ light_ crra`, `monotoneOn_ olgCapital_ light_ crra`, `olgEquilibriumRate_ unique_ light_ crra`	OLG/Light Induction
The guaranteed asset path	`assetFloor`, `assetFloor_ le_ stagePolicy`	OLG/Asset Incidence
Human wealth with assets in it	`assetHumanWealth`, `assetHumanWealth_mono`	OLG/Asset Incidence
Theorem 7.11	`assetHumanWealth_ le_ stageConsumption`	OLG/Asset Incidence

Result	Lean name	Module
It dominates the asset-free bound	riskHumanWealth_le_assetHumanWealth	OLG/Asset Incidence
Proposition 6.14	cohortAssets_le_of_incidence, olgCapital_le_of_incidence	OLG/Existence
Existence with the sharper ceiling	exists_olgEquilibrium_of_sharpBounds	OLG/Existence
Proposition 6.15	cohortCeil, cohortAssets_le_cohortCeil	OLG/Existence
Ceiling on aggregate capital	olgCapital_le_cohortCeil	OLG/Existence
Theorem 6.17	exists_olgEquilibrium_affine	OLG/Existence
Proposition 6.12	cohortFloorAt, cohortFloorAt_le_cohortAssets	OLG/Existence
Floor on capital, clamped	le_olgCapital_at	OLG/Existence
Theorem 6.18	exists_olgEquilibrium_clamped	OLG/Existence
The summed pairing	summedPairing, SummedLasryLionsMonotone	Equilibrium/ LasryLions
Theorem 9.1	summedPairing_nonpos_of_pathEquilibrium	Equilibrium/ LasryLions
Date-by-date is a special case	summedLasryLionsMonotone_of_forall, summedPairing_eq_zero_of_pathEquilibrium	Equilibrium/ LasryLions
Theorem 9.2	aggregate_eq_of_pathEquilibrium	Equilibrium/ LasryLions
Costs seeing the whole family	famPathCost, IsFamPathEquilibrium, famSummedPairing	Equilibrium/ LasryLions
The engine in that generality	famSummedPairing_nonpos_of_pathEquilibrium	Equilibrium/ LasryLions
Theorem 9.3	famLifetime, famSummedPairing_scalar, statistic_eq_of_increasingDifferences	Equilibrium/ LasryLions
Proposition 9.5	famCost, famSummedPairing_eq_neg_losses	Equilibrium/ LasryLions
Each cross-loss is nonnegative	famCost_le_of_pathEquilibrium, famSummedPairing_nonpos_of_losses	Equilibrium/ LasryLions
Proposition 9.6	monotone_of_comonotone_of_strictMono	Equilibrium/ LasryLions
The accumulated gap	gapAccum, gapAccum_nonneg	OLG/Increment
Theorem 10.1	negPart_le_gapAccum, neg_gapAccum_le	OLG/Increment
Aggregate form, single crossing	sum_neg_gapAccum_le, lt_of_increment_bound	OLG/Increment
Superadditive certainty equivalent	powerMean_add_const_le	OLG/Incidence
Theorem 10.2	stageMPC_mul_le_stageConsumption_sub	OLG/Propensity Floor
Theorem 10.3	cohortAssets_sub_le_cohortSlope	OLG/Propensity Floor
Theorem 10.4	incomeStep, stageStep_of_incomeFn, sum_mul_incomeStep	OLG/Mass Weighted

Result	Lean name	Module
The marginal at every age	`sum_mul_incomeStep_iterate`, `sum_mul_le_of_le`	OLG/Mass Weighted
Theorem 1 from the CE criterion	`saving_nonneg_of_correction_le`	OLG/Certainty Equivalent
The precautionary split	`saving_le_of_precautionary_fall`	OLG/Certainty Equivalent
Level and ratio routes	`saving_le_of_precautionary_level`, `saving_le_of_precautionary_ratio`	OLG/Certainty Equivalent
Level route from a floor	`precautionary_level_of_floor`	OLG/Certainty Equivalent
Proposition 7.4	`le_of_ceiling_le_floor`	OLG/Certainty Equivalent
Proposition 10.5	`mul_le_mul_of_euler_pair`	OLG/Certainty Equivalent
The damped multiplier	`damped_le_one`	OLG/Certainty Equivalent
The gradient sum, and its warning	`sum_mul_rpow_pred_le`	Analysis/Power Mean
Proposition 10.6	`sum_mul_sq_eq_one_add_variance`	Analysis/Power Mean
Proposition 10.7	`sum_mul_rpow_le_rpow_sum_mul_sq`	Analysis/Power Mean